

INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT

AND

INTERNATIONAL DEVELOPMENT ASSOCIATION

PROJECT APPRAISAL DOCUMENT

ON A

PROPOSED LOAN

IN THE AMOUNT OF US\$240 MILLION

TO THE

ISLAMIC REPUBLIC OF PAKISTAN

FOR A

SECOND KARACHI WATER AND SEWERAGE SERVICES IMPROVEMENT PROJECT

AS PHASE 1 OF THE MULTI-PHASE PROGRAMMATIC APPROACH
PAKISTAN URBAN WATER, SANITATION, AND HYGIENE (WASH) SERVICES PROGRAM

WITH AN OVERALL FINANCING ENVELOPE OF US\$840 MILLION

November 18, 2024

Water Global Practice
South Asia Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective September 30, 2024)

Currency Unit = Pakistan Rupee (PKR)

PKR 277.6998 = US\$1

US\$0.0036 = PKR 1

FISCAL YEAR

July 1 to June 30

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ABBREVIATIONS AND ACRONYMS

ADB	Asian Development Bank
AIIB	Asian Infrastructure Investment Bank
AM	Accountability Mechanism
Cat DDO	Catastrophe-Deferred Drawdown Option
CCDR	Country Climate and Development Report
CERC	Contingent Emergency Response Component
CHS	Community Health and Safety
CO ₂	Carbon Dioxide
CoC	Commitment of Cooperation
CPF	Country Partnership Framework
CPS	Country Partnership Strategy
DA	Designated Account
E&S	Environmental and Social
ERs	Emission Reductions
ESCO	Energy Service Company
ESCP	Environmental and Social Commitment Plan
ESF	Environmental and Social Framework
ESIA	Environmental and Social Impact Assessment
ESMP	Environmental and Social Management Plan
ESS	Environmental and Social Standards
FM	Financial Management
FY	Fiscal Year
GAP	Gender Action Plan
GBV	Gender-Based Violence
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GIS	Geographic Information System
GoP	Government of Pakistan
GoS	Government of Sindh
GRM	Grievance Redress Mechanism
GRS	Grievance Redress Service
IBRD	International Bank for Reconstruction and Development
IDA	International Development Association
IFC	International Finance Corporation
ILC	In-Line Chlorination
IMF	International Monetary Fund
IPF	Investment Project Financing
IRR	Internal Rate of Return
IUFR	Interim Unaudited Financial Reports

KBFU	Kalri Baghar Feeder Upper
km	Kilometers
KTP	Karachi Transformation Plan
KWSC	Karachi Water & Sewerage Corporation
KWSSIP	Karachi Water and Sewerage Services Improvement Project
LARP	Land Acquisition and Resettlement Plan
M&E	Monitoring and Evaluation
MDB	Multilateral Development Bank
MGD	Million Gallons per Day
MIS	Management Information System
MPA	Multiphase Programmatic Approach
NPV	Net Present Value
NRW	Non-Revenue Water
O&M	Operations and Maintenance
OCCR	Operating Cost Coverage Ratio
OHS	Occupational Health and Safety
OP	Operational Policy
PCP	Punjab Cities Program
PICP	Punjab Inclusive Cities Program
PDO	Project Development Objective
PforR	Program-for-Results
PIE	Project Implementing Entity
PIU	Project Implementation Unit
PKR	Pakistani Rupee
PMC	Program Management Consultancy
PPP	Public-Private Partnership
PPSD	Project Procurement Strategy for Development
PrDO	Program Development Objective
PSC	Project Steering Committee
SEA	Sexual Exploitation and Abuse
SMF	Social Management Framework
US\$/USD	United States Dollar
VGF	Viability Gap Fund
WAPDA	Water and Power Development Authority
WASH	Water, Sanitation, and Hygiene
Water GCP	Water Security and Climate Adaptation Global Challenge Program
WBG	World Bank Group
WSS	Water Supply and Sanitation

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**DATASHEET****BASIC INFORMATION**

Project Beneficiary(ies)	Operation Name		
Pakistan	Second Karachi Water and Sewerage Services Improvement Project (KWSSIP-2)		
Operation ID	Financing Instrument	Environmental and Social Risk Classification	
P171422	Investment Project Financing (IPF)	High	

Financing & Implementation Modalities

☒ Multiphase Programmatic Approach (MPA)	☐ Contingent Emergency Response Component (CERC)
☐ Series of Projects (SOP)	☐ Fragile State(s)
☐ Performance-Based Conditions (PBCs)	☐ Small State(s)
☐ Financial Intermediaries (FI)	☐ Fragile within a non-fragile Country
☐ Project-Based Guarantee	☐ Conflict
☐ Deferred Drawdown	☐ Responding to Natural or Man-made Disaster
☐ Alternative Procurement Arrangements (APA)	☐ Hands-on Expanded Implementation Support (HEIS)

Expected Approval Date	Expected Closing Date	Expected Program Closing Date
12-Dec-2024	31-Dec-2030	31-Dec-2034
Bank/IFC Collaboration	Joint Level	
Yes	Complementary or Interdependent project requiring active coordination	

MPA Program Development Objective

To provide safely managed WASH services in urban areas in Pakistan

MPA FINANCING DATA (US\$, Millions)



MPA Program Financing Envelope	2,180.00
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Components

Component Name	Cost (US\$)
Component 1: Capacity Building and Reform of KWSC	14,000,000.00
Component 2: Safely Managed Water Supply and Sanitation Services	958,560,000.00
Component 3: Project Management and Studies	25,240,000.00
Component 4: Contingent Emergency Response Component (CERC)	0.00

Organizations

Borrower: Islamic Republic of Pakistan

Implementing Agency: Province of Sindh through the Karachi Water and Sewerage Corporation (KWSC)

MPA FINANCING DETAILS (US\$, Millions)

MPA Program Financing Envelope:	2,180.00
of which Bank Financing (IBRD):	440.00
of which Bank Financing (IDA):	400.00
of which Other Financing sources:	1,340.00

PROJECT FINANCING DATA (US\$, Millions)**Maximizing Finance for Development**

Is this an MFD-Enabling Project (MFD-EP)? No

Is this project Private Capital Enabling (PCE)? Yes

SUMMARY

Total Operation Cost	999.00
Total Financing	999.00
of which IBRD/IDA	240.00



Financing Gap

0.00

DETAILS

Private Sector Investors/Shareholders

Equity	Amount	Debt	Amount
Government Contribution	730.00	Commercial Debt	188.00
Government Resources	250.00	Unguaranteed	188.00
IBRD	240.00		
Other Donors	240.00		
Non-Government Contribution	81.00		
Private sector Equity	81.00		
Total	811.00		188.00

Expected Disbursements (US\$, Millions)

WB Fiscal Year	2025	2026	2027	2028	2029	2030	2031
Annual	9.33	24.00	36.00	48.00	48.50	50.00	24.17
Cumulative	9.33	33.33	69.33	117.33	165.83	215.83	240.00

PRACTICE AREA(S)

Practice Area (Lead)

Water

Contributing Practice Areas

Urban, Resilience and Land; Infrastructure, PPP's & Guarantees

CLIMATE

Climate Change and Disaster Screening

Yes, it has been screened and the results are discussed in the Operation Document



SYSTEMATIC OPERATIONS RISK- RATING TOOL (SORT)

Risk Category

Rating

1. Political and Governance	● High
2. Macroeconomic	● High
3. Sector Strategies and Policies	● Moderate
4. Technical Design of Project or Program	● Moderate
5. Institutional Capacity for Implementation and Sustainability	● Substantial
6. Fiduciary Financial Management Risk rating from Specialist: ● Moderate as of 2024-07-31T07:20:10Z Procurement Risk rating from Specialist: ● Substantial as of 2024-07-29T00:00:00Z	● Substantial
7. Environment and Social Environment Risk rating from Specialist: ● Substantial as of 2024-08-26T10:18:17Z Social Risk rating from Specialist: ● High as of 2024-08-26T10:18:17Z	● High
8. Stakeholders	● Substantial
9. Other	● Substantial
10. Overall	● Substantial
Overall MPA Program Risk	● Substantial

POLICY COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?

☐ Yes ☒ No

Does the project require any waivers of Bank policies?

☐ Yes ☒ No

ENVIRONMENTAL AND SOCIAL



Environmental and Social Standards Relevance Given its Context at the Time of Appraisal

E & S Standards	Relevance
ESS 1: Assessment and Management of Environmental and Social Risks and Impacts	Relevant
ESS 10: Stakeholder Engagement and Information Disclosure	Relevant
ESS 2: Labor and Working Conditions	Relevant
ESS 3: Resource Efficiency and Pollution Prevention and Management	Relevant
ESS 4: Community Health and Safety	Relevant
ESS 5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement	Relevant
ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources	Relevant
ESS 7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities	Not Currently Relevant
ESS 8: Cultural Heritage	Not Currently Relevant
ESS 9: Financial Intermediaries	Not Currently Relevant

NOTE: For further information regarding the World Bank's due diligence assessment of the Project's potential environmental and social risks and impacts, please refer to the Project's Appraisal Environmental and Social Review Summary (ESRS).

LEGAL

Legal Covenants

Sections and Description

1. The Project Implementing Entity (PIE) shall, no later than twelve (12) months from the Effective Date, enter into a performance agreement with KWSC, which shall include (a) performance targets for KWSC and action steps for regular performance review by the PIE; and (b) updates and revisions to the principles and objectives set out in the Commitment of Cooperation, all in form and substance acceptable to the Bank.
2. The PIE shall, no later than six (6) months after the Effective Date, appoint a performance auditor acceptable to the Bank to monitor and carry out performance assessments of KWSC's achievement of the targets set out in the performance agreement between the PIE and KWSC.
3. The PIE shall, no later than eighteen (18) months from the Effective Date, publish the performance auditor's first annual performance assessment report of KWSC, and shall provide the Bank for its review the results of the performance auditor's annual performance assessments no later than six months after the end of the Fiscal Year, and promptly thereafter disclose those assessments to the public in the form of annual performance reports of KWSC.
4. The PIE shall cause KWSC to prepare, within one (1) year from the Effective Date, a financial sustainability roadmap, acceptable to the Bank, based on a utility financial model with time-bound actions to achieve an OCCR for water supply



operations of at least one hundred thirty percent (130%) (including electricity costs and without compensation of these costs).

5. The PIE shall take all necessary measures to ensure that KWSC's total revenues are not less than the sum of its total operating expenses and debt service requirements (if any).

6. The PIE shall cause KWSC to have its financial statements for each Fiscal Year audited by independent auditors acceptable to the Bank in accordance with consistently applied accounting standards acceptable to the Bank and make such audited financial statements publicly available in a timely fashion in a manner acceptable to the Bank.

7. The PIE shall, no later than June 30, 2027, develop operating rules for the Keenjhar Lake based on a revised water allocation model to ensure at least ninety-four percent (94%) supply reliability to Karachi city.

8. The PIE shall, no later than eighteen (18) months from the Effective Date, adopt a new water and wastewater tariff review framework that enables recovery of the full cost of service provision on a timeline acceptable to the Bank.

9. Without limitation to the provisions of Section 5.03 of the General Conditions, the Borrower shall cause the PIE to provide an amount at least equivalent to one hundred twenty million Dollars (\$120,000,000) as the Counterpart Funds for the Project.

Conditions

Type	Citation	Description	Financing Source
Effectiveness	5.01 (a)	The PIU has recruited one senior environmental specialist and one senior social development specialist.	IBRD/IDA
Effectiveness	5.01 (b)	The POM has been updated, in form and substance acceptable to the Bank, and adopted by KWSC.	IBRD/IDA
Effectiveness	5.01 (c)	The K-IV Coordination and Environmental and Social (E&S) Implementation Committee described in the ESCP has been formed.	IBRD/IDA
Effectiveness	5.01 (d)	The AIIB Co-financing Agreement has been executed and delivered and all conditions precedent to its effectiveness or to the right of the Borrower to make withdrawals under it (other than the effectiveness of the Loan	IBRD/IDA



		Agreement) have been fulfilled.	
Disbursement	S2. III. B.1. (a)	Notwithstanding the provisions of S.2.III.A of the Loan Agreement, no withdrawal shall be made for payments made prior to the Signature Date, except that withdrawals up to an aggregate amount not to exceed (i) forty-three million five hundred thousand Dollars (\$43,500,000) may be made for payments made prior to this date but on or after November 1, 2024, for Eligible Expenditures under Category (1) and (ii) four million five hundred thousand Dollars (\$4,500,000) may be made for payments made prior to this date but on or after November 1, 2024, for Eligible Expenditures under Category (2)(a), provided that the Borrower has completed, in a manner and substance satisfactory to the Bank, the environmental and social audit referred to in the ESCP for those works under Part 2.5(i) of the Project completed prior to the request for such withdrawal	IBRD/IDA
Disbursement	S2. III. B.1. (b)	Notwithstanding the provisions of S.2.III.A of the Loan Agreement, no withdrawal shall be made for goods, works, non-	IBRD/IDA



		consulting services, and consulting services under Part 2.5(ii) of the Project under Category (2)(b), unless and until: (i) in the Bank's opinion at the time of contract award for Part 2.5(ii) (or such other time as the Bank may determine in its sole discretion), K-IV Phase 1 and works to deliver increased annual inflows to Keenjhar Lake by an amount greater than or equal to the required amount have been or will, at the same time as the activities contemplated under Part 2.5(ii) of the Project, be completed to the satisfaction of the Bank; and (ii) the corrective actions needed to align the K-IV Phase 1 associated facilities with the requirements of the World Bank Environmental and Social Framework have been implemented and completed in a manner acceptable to the Bank	
Disbursement	S2. III. B.1. (c)	Notwithstanding the provisions of S.2.III.A of the Loan Agreement, no withdrawal shall be made for Emergency Expenditures under Category (6), unless and until all of the following conditions have been met in respect of said expenditures: (i)(A) the Borrower has determined	IBRD/IDA



		that an Eligible Crisis or Emergency has occurred, and has furnished to the Bank a request to withdraw Loan amounts under Category (6); and (B) the Bank has agreed with such determination, accepted said request and notified the Borrower thereof; and (ii) the Borrower has adopted the CERC Manual and Emergency Action Plan, in form and substance acceptable to the Bank.	
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I. STRATEGIC CONTEXT

A. Country Context

- Poverty has increased slightly amid recent shocks, despite some economic stabilization.** Pakistan made significant progress towards reducing poverty between 2001 and 2018 with the expansion of off-farm economic opportunities and increased external remittances. However, this has not fully translated into improved socio-economic conditions: over one-third of school-age children across Pakistan were found to be out of school; nearly two-thirds of those in school in fiscal year (FY)24 were learning deprived; and alarmingly high rates of stunting (40 percent in FY23) persist. Critical constraints, including recurrent fiscal and current account deficits, protectionist trade policies, unproductive agriculture, a difficult business environment, a heavy state presence in the economy, and a financially unsustainable energy sector, have remained largely unaddressed, leading to slow and volatile growth. Amid the COVID-19 pandemic, the catastrophic 2022 floods and macroeconomic volatility, poverty has increased. The estimated lower-middle income poverty rate is 40.5 percent (US\$3.65/day 2017 purchasing power parity) for FY24 with an additional 2.6 million Pakistanis falling below the poverty line from the year before.
- Pakistan experienced heavy monsoon rains in 2022 leading to catastrophic and unprecedented flooding with enormous human and economic impacts.** Roughly 33 million people were impacted, and many permanently displaced. More than 13,000 kilometers (km) of roads were destroyed, 2.2 million houses damaged, around 3.8 million hectares of crops were flooded, and an estimated 1.2 million livestock were killed. Limited access to input and output markets and temporary disruptions to supply chains subsequently drove up food prices and added to existing price pressures resulting from reduced agricultural yields and the global rise of food prices. The government's Post-Disaster Needs Assessment¹ estimated that the need for rehabilitation and reconstruction is at US\$16.3 billion.
- Pakistan has made recent progress towards macroeconomic stabilization, but risks remain extremely high and faster sustained growth will require substantial reform.** At the beginning of FY24, Pakistan's economy faced a potential economic crisis in the face of political uncertainty, global monetary policy tightening, and fiscal and external imbalances, that led to pressures on domestic prices and foreign reserves. To preserve reserves, measures to manage imports and capital outflows were introduced, which disrupted local supply chains, economic activity and exacerbated inflationary pressures. Under the interim government, an International Monetary Fund (IMF) Stand-By Arrangement was approved in July 2023. Consequently, exchange rate flexibility was restored, import controls were relaxed, and steps were taken to contain the fiscal deficit. Political uncertainty also diminished with the successful conduct of the general elections. Coupled with favorable weather conditions and easing external conditions, the economy began recovering in FY24. Consequently, growth of real gross domestic product (GDP) at factor cost is estimated to have risen to 2.5 percent year on year in FY24, after contracting by 0.2 percent in FY23. Downside risks remain high, with the outlook predicated on the new IMF Extended Fund Facility program remaining on track, continued fiscal restraint, and additional external financing. Heavy banking sector exposure to the sovereign, domestic policy uncertainty, and geopolitical instability pose significant risks to the outlook. Robust economic recovery over the medium term will require the steadfast implementation of much broader fiscal and economic reforms.

¹ Government of Pakistan. 2022. *Pakistan Floods 2022 Post-Disaster Needs Assessment*. Ministry of Planning Development & Special Initiatives.



4. **According to the 2022 Country Climate and Development Report (CCDR), Pakistan's high vulnerability to climate change is a risk multiplier, compounding its human and economic development challenges.** Pakistan consistently ranks among the top 10 countries most affected by climate change worldwide.² More frequent and extreme weather events adversely impact ecosystems, people, and their settlements and infrastructure. Droughts, flooding, and heat waves are prevalent risks. Coastal cities such as Karachi are also vulnerable to climate change-induced sea level rise, cyclones, saltwater intrusion, and coastal erosion. The risks from climate change and environmental degradation could further aggravate Pakistan's economic fragility and ultimately reduce annual GDP by 18 to 20 percent per year by 2050.

5. **Pakistan also faces an unfinished agenda on human development, which compounds its structural macro fragility and creates a drag on the economy.** The country currently ranks 141 out of 174 on the Human Capital Index, scoring 41 out of 100. The stunting rate of children under five remains among the highest in the world at 40 percent. Evidence suggests that the lack of safely managed water supply and sanitation and basic hygiene (WASH) services is partly responsible for this staggering loss of human potential.³

B. Sectoral and Institutional Context

6. **Safely managed WASH services are the foundation of public health and quality of life and are central to addressing the stunting crisis in Pakistan. However, the country lags on WASH service indicators because of decades of underinvestment.** Close to 126 million people (half of the population) do not receive safely managed water services, 166 million (three-quarters of the population) do not have access to safely managed sanitation services, and 36 million lack basic hygiene services. Data from the Pakistan Council of Research in Water Resources showed that 61 percent of drinking water samples tested nationwide failed to meet national drinking water quality standards in 2020.⁴

7. **WASH is a provincial mandate, and provinces are at different stages in addressing service gaps.** Punjab and Sindh, the two largest provinces, have made commitments to address service gaps and have enabling policies and plans in place, such as the Punjab Water Policy 2018, Punjab WASH Sector Development Plan (2014-2024), Sindh Water Policy 2023, and Sindh WASH Sector Development Plan (2016-2026). Both provinces have also passed laws to enable service providers to become financially sustainable. In Punjab's secondary cities, WASH services are the mandate of local governments. The Punjab Water Act (2019) strengthened the regulatory framework for water service delivery. Similarly, the recently enacted Karachi Water and Sewerage Corporation Act, 2023 converted the Karachi Water and Sewerage Board into a corporate entity with enhanced financial and managerial autonomy. However, massive capital injection is needed to address infrastructure gaps in all provinces.

8. **Traditional investment approaches have been too small and fragmented.** The 2014 Punjab WASH Sector Development Plan identified the need for annual spending of PKR 40 billion to provide two-thirds of the population with safely managed WASH services by 2024. However, actual spending was insufficient and focused mainly on infrastructure, neglecting operation and maintenance (O&M) and service delivery. Sindh did not sufficiently finance its WASH Sector Development Plan (2016-2026), and service indicators have continued to worsen. Low and unpredictable budget have led to ad hoc engagements. The absence of systematic monitoring undermined accountability and limited the ability of

² Eckstein, David, Vera Künzel, and Laura Schäfer. 2021. "Global Climate Risk Index 2021." Germanwatch Briefing Paper.

³ Mansuri, Ghazala et al. 2018. *When Water Becomes a Hazard: A Diagnostic Report on The State of Water Supply, Sanitation and Poverty in Pakistan and Its Impact on Child Stunting*. WASH Poverty Diagnostic Washington, D.C.: World Bank Group.

⁴ Hifza, R., Fauzia, A., Kiran, A., and M. Ashraf. 2021. "Drinking Water Quality in Pakistan: Current Status and Challenges." Pakistan Council of Research in Water Resources, Islamabad.



governments to amend plans. Provincial plans need more credible financing mechanisms.

9. **A new generation of provincial programs is emerging that places more emphasis on service delivery and the financial performance of service providers.** The Punjab Cities Program (PCP) is building the capacity of local governments to provide WASH services in selected secondary cities⁵ and improving revenue collection and O&M systems. It has reached 3.15 million people (about 79 percent of the program area's population) with safely managed WASH services. In Sindh, the 2019 Karachi Transformation Plan (KTP), a joint program of the Government of Sindh (GoS) and the Government of Pakistan (GoP), promised to address service delivery issues, including WASH, by investing in infrastructure and transforming the performance of service providers such as the Karachi Water and Sewerage Corporation (KWSC).

10. **The KWSC is the largest water utility in Pakistan, and, given the size and role of Karachi as Pakistan's economic and financial hub, its performance is of national significance.** The KWSC provides WASH services to about 17 million people, almost 84 percent of Karachi's population. Service hours are intermittent and range from two hours every two days to four hours per day with widespread rationing. Many *Katchi Abadis*⁶ are unconnected and rely on informal water sources. Non-revenue water (NRW) is around 64 percent in part due to insufficient hydraulic capacity and leakages. The sewer network covers only about 75 percent of the city, with aging pipes, insufficient capacity, malfunctioning pumping facilities, and a lack of treatment capacity. The city's three wastewater treatment plants operate at one-third of capacity, treating 50 million gallons per day (MGD) of the estimated 472 MGD generated.

11. **Karachi and the KWSC's water supply challenges are compounded by high and increasing competition for water (especially for Indus River water), a deteriorated and complex bulk water supply system, and the lack of regional water resources planning.** Karachi's primary water source is the Indus River, which also supplies the province's extensive irrigation system and drinking water outside of Karachi. Currently, Karachi faces a water supply deficit of about 580 MGD,⁷ but increasing supply requires investment and coordination. However, there is no water resources management plan or allocation mechanism within Sindh, nor are there formal operating rules for critical infrastructure conveying water to Karachi, which is mostly managed by the Sindh Irrigation Department.

12. **From a financial and human resources standpoint, KWSC performs well below expectations for a water utility serving a major commercial hub.** The utility has 1.07 million registered customers compared to Karachi Electric's 3.5 million in the same area. No household has a meter, and only 75 percent receive bills. Billing efficiency is about 36 percent, and O&M cost recovery is around 73 percent. The total number of staff is almost 10,000, but staff productivity is low at 9.5 per 1,000 connections. At the same time, the utility has been unable to recruit new talent due to a GoS hiring moratorium from 2008 to 2024. A lack of diversity in the workforce undermines the KWSC's potential. Women comprise only 4 percent of the workforce and only 10 women hold technical and managerial positions compared to 630 men. Due to the water sector being traditionally male-dominated and perceived as not supportive of women, as well as a lack of skills training, it is difficult for women to break into the sector. Recently, the KWSC has been making efforts to address its gender gap by establishing a Women's Development Centre that includes a daycare and increasing training opportunities for women employees.

13. **In 2019, as part of the KTP, the GoS embarked on a 10-year investment program to provide universal access to safely managed WASH services in Karachi, backed by reform commitments to**

⁵ In this context, secondary cities exceed the 100,000 population threshold, excluding provincial and national capitals. World Bank. 2024. *Realizing the Potential of Pakistan's Secondary Cities*. Washington, DC: World Bank Group.

⁶ *Katchi Abadis* are unplanned settlements as declared by the Sindh Katchi Abadis Authority.

⁷ The per capita demand used to calculate the deficit is 40 gallons per capita per day for domestic use only.



transform the KWSC into a financially viable utility. The investment program addresses five structural problems in Karachi's water and sanitation system: bulk water supply shortfall, a leaky water distribution network, an unreliable supply that does not meet quality standards, a lack of wastewater collection and treatment, and persistent poor service levels, especially in Katchi Abadis. The program included short-term interventions to rapidly improve services and KWSC revenues while laying the groundwork for more significant infrastructure investments. The World Bank supported the program by signing a Commitment of Cooperation (CoC) for the reform⁸ of the KWSC together with the GoS and through a US\$40 million International Bank for Reconstruction and Development (IBRD) loan for the Karachi Water and Sewerage Services Improvement Project (KWSSIP-1; P164704). Additionally, KWSC completed a Utility Performance Improvement Assessment, and Training for key staff focused on customer service management and billing development with Aguas de Portugal, concluding in June 2024.

14. The KWSC is on track to deliver, for the first time, safe drinking water for 2 million people by June 2025. The 2023 KWSC Act strengthened the KWSC's financial and managerial autonomy. A new Board of Directors with more independent members appointed a new KWSC Chief Executive Officer and Chief Operating Officer in 2023 through open market competition. The new management has prioritized revenue-enhancing measures, including digitalizing hydrant management, addressing illegal connections, increasing metering, and recovering arrears. As a result of these measures, the KWSC's revenue has more than doubled from PKR 8,305 million in FY19 to PKR 16,428 million in FY23.⁹ Construction is underway of the Greater Karachi Bulk Water Supply Scheme (K-IV) Phase 1 to increase KWSC's bulk water capacity. KWSC is also making investments to control water losses and developing two public-private partnerships (PPPs) for wastewater treatment and recycling (TP-I and TP-IV) advised by the Asian Development Bank (ADB). TP-I is at the Request for Proposal stage and is expected to reach financial close by July 2025.

15. The new generation of WASH programs in Sindh and Punjab is a springboard for new national WASH platforms to leverage capital from development partners and the private sector to scale up safely managed WASH services through long-term programmatic infrastructure investments and reforms. The proposed Pakistan Urban WASH Services Multiphase Programmatic Approach (MPA) provides the entry point for such a multi-province platform targeting urban and peri-urban areas. The Second Karachi Water and Sewerage Services Improvement Project (KWSSIP-2) is the proposed first phase of this MPA.

C. Relevance to Higher Level Objectives

16. The MPA is part of a World Bank South Asia regional initiative or program of programs to provide WASH services to about 100 million people across the region by 2035. The proposed Pakistan Urban WASH Services MPA and the Bangladesh Water Security and Water Supply and Sanitation (WSS) MPA are the first in the program. These will likely be followed by a WASH MPA for India in FY25, and a Pakistan Rural WASH services MPA in FY26. Depending on client demand, MPAs or Series of Projects for Nepal and Sri Lanka may also be considered.

17. KWSSIP-2 is consistent with the World Bank Group (WBG) Country Partnership Strategy (CPS) FY15–19 for the Islamic Republic of Pakistan discussed by the WBG Board of Executive Directors on May 1, 2014 (Report No. 84645-PK) and extended to FY20 by the corresponding Performance and Learning Review dated May 18, 2017 (Report No. 113574). The preparation of the new Country Partnership Framework (CPF) was deferred in FY21 due to the COVID-19 crisis and paused in FY23 due to the 2022 floods. The new CPF is expected to be presented to the Board in FY25. The FY15–19 CPS objectives remain

⁸ The Commitment of Cooperation (CoC) for the reform of the KWSC to Improve Service Delivery and Financial Viability of the KWSC was signed between GoS and the World Bank Group on May 30, 2019.

⁹ Data from KWSC Financial Statements.



relevant, and this program contributes to Result Area 4: "Improving Service Delivery." It will lay out a longer-term engagement focusing on a select set of Pakistan's most daunting development challenges, including reducing child stunting through investments in safely managed WASH services.

18. **The program is consistent with and supports Pakistan's climate change strategies**, including the National Water Policy (2018), the National Climate Change Policy (2021), the National Adaptation Plan (2022), and the Nationally Determined Contribution (2021), as well as the Pakistan CCDR. The CCDR calls for strengthening human capital to achieve sustained and equitable development, and climate resilience. It recommends urgent public investment to expand access to safely managed WASH services and investment in climate-resilient water and wastewater infrastructure and services as a priority to transition to a green and resilient economy. Phase 1 will generate emission reductions (ERs) through adopting renewable energy, improvements in energy efficiency, and methane reduction in wastewater collection and treatment. The ERs generated through 2030 are expected to be monetized by the Transformative Carbon Asset Facility as a standalone activity. Additional monetization of ERs from other MPA Phases could be considered. The program is consistent with provincial policies and strategies on urban WASH services and water resources management. Finally, the program aims to mobilize private capital by supporting reforms and improving utility efficiency, financial performance, and creditworthiness.

19. **The MPA is fully aligned with the global framework of the Water Security and Climate Adaptation Global Challenge Program (Water GCP), a priority program in the South Asia Region.** More specifically, the MPA will contribute to achieving beneficiary targets under the Water GCP strategy pillar 1: achieving universal access to safe WSS. The MPA in Pakistan is envisaged as one of the first programs in the South Asia Region to implement the Water GCP and enhance scale, speed, and impact.

D. Multiphase Programmatic Approach

20. **An MPA is conducive to the sustained focus and coordinated investments over a prolonged period at the province level that are needed to address urban WASH services gaps.** It offers several advantages over a standalone Investment Project Financing (IPF) or Program-for-Results (PforR) operation. The MPA assures governments and stakeholders of the World Bank's long-term commitment, offering more predictability to other development partners, including multilateral development banks (MDBs) and the private sector, and providing a platform to align partners, evaluate impact, and share lessons learned across provinces. At the same time, it offers sufficient flexibility to course-correct and adjust the program and its ambition based on results achieved, evolving government priorities, and finances. This Project Appraisal Document introduces the MPA and focuses on KWSSIP-2 as the first phase, signaling the commitment by the World Bank to support critical WASH outcomes over the long term.

21. **The MPA Program Development Objective (PrDO) is to provide safely managed WASH services in urban areas in Pakistan. The PrDO result indicator is "people provided with safely managed WASH services," and the target is 33.5 million by 2035.** The people who count against this indicator are those receiving the three services of safely managed water, safely managed sanitation, and basic hygiene services as defined in the World Bank Corporate Scorecard. Safely managed water means drinking water from an improved water source accessible on premises, available when needed (a minimum of 12 hours per day), and free from fecal and hazardous chemical contaminants (including fluoride, arsenic, and lead). Safely managed sanitation means using an improved sanitation facility not shared with other households and where excreta is safely disposed of in situ or removed and treated offsite. Basic hygiene service means a handwashing facility with soap and water is available at home. The MPA's theory of change is shown in Table 2. Participating provinces will have the flexibility to select from the menu of activities based on their needs, priorities, and capabilities, provided they are aligned with the PrDO and results chain.



Table 1: Proposed Pakistan Urban WASH Services MPA Framework and Indicative Costing (FY25–FY35)

Phase	Phase's Proposed Development Objective	IPF or PforR	IDA/IBRD (US\$ million)	Other (US\$ million)	Estimated Approval Date	Estimated E&S Risk Rating
Phase 1: KWSSIP-2 (P171422)	Provide safely managed water and sanitation services in Karachi and increase KWSC's financial and operational performance.	IPF	240	759	FY25	High
Phase 2: PICP (P508094)	Provide safely managed water and sanitation and basic hygiene services in selected cities of Punjab.	Hybrid PforR	400	80	FY26	Substantial
Phase 3: KWSSIP-3	Provide safely managed water, sanitation and basic hygiene services in Karachi and increase KWSC's financial and operational performance.	IPF	200	501	FY28	High
Total			840	1,340		

22. **The 10-year MPA will initially have three overlapping phases that are expected to scale-up existing successful operations with a total estimated envelope of US\$2.18 billion, including US\$840 million from IBRD and the International Development Association (IDA) and US\$1.34 billion expected from provincial governments, MDBs, and the private sector.** The program is currently expected to target urban areas in Sindh and Punjab. Additional phases could include other provinces or cities, depending on client demand and investment needs. MDBs that have indicated a commitment to the MPA currently include the Asian Infrastructure Investment Bank (AIIB) and the ADB.¹⁰ The European Investment Bank has also expressed potential interest in parallel financing. Private capital will come primarily from equity and debt financing in Karachi (Phases 1 and 3) and potentially in Punjab Inclusive Cities Program (PICP, as the second phase of PCP). Table 1 summarizes the program plan.

23. **Phase 1 will build up on KWSSIP-1 to provide 6 million people in Karachi with all three safely managed WASH services.** It will leverage GoS and GoP capital already committed under the KTP, as well as MDB and private sector capital, to expand the scope of investments in bulk water augmentation, water treatment, wastewater treatment and reuse, behavior change, water distribution, and sewer network rehabilitation, including in Katchi Abadis. This phase will also deepen support for KWSC reform.

24. **Phase 2, which is currently under discussion, intends to target 20 secondary cities in Punjab to reach 11.5 million people with all three safely managed WASH services, building on and expanding the scope of the successful ongoing PCP in the province.** It is proposed to target secondary cities where child stunting is highest and partner with local governments to expand access to safely managed water and sanitation services, promote hygiene behavior change, and improve solid waste management, drainage, and flood management. PICP proposes to strengthen the institutional capacity of local governments by modernizing systems that support sustained, safely managed WASH service delivery, such as O&M and asset management, water quality monitoring, billing, revenue collection, and financial management.

¹⁰ AIIB is co-financing Phase 1 with US\$240 million. The World Bank will assume full fiduciary responsibility and operate within the cofinancing policy for cost recovery in accordance with a co-lending agreement with AIIB. ADB is transaction the advisor for TP-IV and is a potential financier of US\$50 million for viability gap financing of TP-IV to the GoS as parallel financing.



Table 2: Theory of Change for the Pakistan Urban WASH Services MPA

Menu of Activities	Typical Outputs	Outcomes	Impacts
Pillar 1: Safely Managed Water Services		People provided with safely managed water services.	Reduced child stunting.
• Last-mile connectivity to water networks • Treatment plant and distribution network rehabilitation and expansion. • Household water connections. • Water quality surveillance. • In-line chlorination (ILC) units and point-of-use treatment systems. • Source sustainability and augmentation. • Communal potable water filtration plants.	• Household piped connections installed. • Schools and health centers provided with WASH. • Network rehabilitated/expanded. • Functional water treatment plants. • ILC and point-of-use units installed. • Water sources augmented. • Communities provided with communal potable water points		
Pillar 2: Safely Managed Sanitation and Basic Hygiene Services		People provided with safely managed sanitation services.	Reduced incidence of water-related diseases.
• Last-mile connectivity to sewer networks • Wastewater recycling plants. • Sewer and stormwater drainage network expansion and rehabilitation. • Household sewer connections. • Non-sewered sanitation and fecal sludge management. • Handwashing facilities with soap and water. • WASH behavior change communication. • Solid waste recovery and sanitary disposal. • Sustainable drainage and solutions for water retention, percolation and storage.	• Sewerage connections installed. • Safe non-sewered sanitation in use. • Wastewater treatment and fecal sludge management plants operational. • Handwashing facilities with soap and water installed and in use. • Increased awareness/adoption of hygiene practices. • Solid waste recovery and sanitary disposal infrastructure operational. • Stormwater drainage and climate adaptation infrastructure operational • Reduced greenhouse gas emissions		
Pillar 3: Institutional Strengthening and Financial Sustainability		People provided with improved stormwater drainage services.	
• Technical, commercial, and operational efficiency interventions. • Capacity building of service providers. • Service performance monitoring. • Utility and institutional/regulatory reforms • Updated WASH programs and financing roadmaps. • PPP market facilitation, preparation, and PPP advisory. • Development of city resilience plans • Revision of rates for key taxes and fees. • Expansion of Urban Immovable Property Tax base. • Accessing eligible carbon financing.	• Performance agreements signed. • Billing/collection efficiency improved. • NRW reduced. • PPP contracts signed. • Service providers’ operating cost coverage ratios increased. • Public financing committed to WASH. • Private capital mobilized for WASH. • Revised regulations, standards and specifications for WASH services • Increased revenues from land and property-related taxes and fees • Improved energy and resource efficiency		
<i>Critical assumptions: (i) households have financial means and willingness to connect to water and sanitation systems; (ii) people reached with hygiene awareness campaigns are the correct target (e.g., women); (iii) people reached have access to quality preventative/curative health care, nutrition, and food security.</i>			



25. **Phase 3 aims to scale up all three safely managed WASH services to 16 million additional people in Karachi**, building on KWSSIP-2. It would leverage GoS, GoP, MDBs, and private sector capital to expand investments in water treatment, wastewater treatment and reuse, behavior change, water distribution, and sewer network rehabilitation and expansion, including in Katchi Abadis. It would scale up metering (district, bulk user, and households), service connection rehabilitation, and telemetry replacement and installation to better monitor and operate WSS systems. It would continue to support KWSC's reform, focusing on access to corporate finance or private capital. Phase 3 may also include a partial risk guarantee to support PPPs, subject to meeting appraisal requirements, including market demand.
26. **The MPA includes a robust learning agenda to enable learning across provinces on WSS service provider reforms and hygiene behavioral change.** The MPA will facilitate knowledge sharing, drawing on country, regional, and global evidence on how different institutional models (corporatized utilities, statutory bodies, ringfenced municipal departments, etc.) perform in various contexts and what reforms are needed. For instance, lessons from utility corporatization reforms in Karachi under Phase 1 will inform the design of similar reforms for water service providers in secondary cities in Punjab under Phase 2. In addition, lessons from PPP transactions and utility turnaround strategies in Phase 1 will inform similar initiatives in Phases 2 and 3. Moreover, the MPA will invest in building institutional capacity for capturing and sharing lessons at the municipal and utility levels so that the learning could, over time, benefit other regions as well. Phase 3 will be informed by the lessons learned from Phases 1 and 2 as well as KWSSIP-1.
27. **The learning agenda will generate evidence to inform context-specific hygiene behavioral change interventions to maximize public health gains from WASH services.** Robust monitoring and evaluation (M&E) of MPA activities will help assess the impact of such interventions on outcomes like reduced exposure to water-borne diseases and child stunting. Program assessments will answer short-run behavior questions, such as the effective framing of hygiene messages to encourage the adoption of new behaviors and the uptake of services. In contrast, more traditional impact evaluations will answer longer-term questions, such as the impact of program interventions on reducing child stunting.
28. **The Bank will facilitate learning through a country WASH knowledge platform.** This platform will deliver a mix of analytical, capacity-building, advisory, and convening services (forums, learning events, networking) to ensure the diffusion of knowledge and learning into each MPA phase. Financing for each phase will cover some aspects of the knowledge platform (e.g., long-term impact evaluations). Others, such as program assessments, capacity building, and advisory, will be Bank-executed. The learning agenda will also draw on the World Bank's published body of knowledge, including case studies on the characteristics of well-performing public water utilities,¹¹ water utility turnaround frameworks,¹² water utility maturity models,¹³ roadmaps for building the resilience of water utilities to climate change,¹⁴ as well as recent work on behavior insights for the water sector.¹⁵
29. **The Bank will leverage existing country and regional networks, including the Pakistan Water**

¹¹ Baietti, Aldo, William Kingdom, and Meike van Ginneken. 2006. "Characteristics of Well-performing Public Water Utilities." Water Supply and Sanitation Working Notes No. 9. World Bank Group, Washington, DC.

¹² Soppe, Gerard, Nils Janson, and Scarlett Piantini. 2018. "Water Utility Turnaround Framework: A Guide for Improving Performance." World Bank Group, Washington, DC.

¹³ Kayaga, Sam Mbaziira, William D. Kingdom, and Josses Mugabi. 2013. "Evaluating the Institutional Sustainability of an Urban Water Utility: A Conceptual Framework and Research Directions." *Utilities Policy* 27 2013-12.

¹⁴ World Bank. 2018. "Building the Resilience of WSS Utilities to Climate Change and Other Threats: A Road Map." World Bank, Washington, DC.

¹⁵ Joseph, George, Sophie Ayling, and Alejandra Quevedo Cardona. 2023. "Behavioral Insights for the Water Sector: Improving Outcomes by Changing Mindsets." World Bank, Washington, DC.



Operators Network, the South Asia Water Utility Hub, and the Singapore Water Center, to provide opportunities for water utilities to regularly share knowledge and experiences on common issues such as tariff reform, human resource reforms, performance management, digitization, NRW reduction, and PPPs.

II. PROJECT DESCRIPTION

A. Project Development Objective

30. **The Project Development Objective (PDO) of Phase 1 is to provide safely managed water and sanitation services in Karachi and increase KWSC's financial and operational performance.** The PDO indicators are: (i) People provided with safely managed water (including female and youth) (number); (ii) People provided with safely managed sanitation (including female and youth) (number); and (iii) KWSC operating cost coverage ratio improved (percent). The first two are WBG Corporate Scorecard indicators.

B. Project Components

31. **The project scope consists of three components that contribute to the PDO** and a Contingent Emergency Response Component (CERC) with zero allocation, providing flexibility to the GoP/GoS to reallocate funds to support emergency response activities in case of an eligible emergency.¹⁶

32. **Component 1 – Capacity Building and Reform of KWSC (US\$14 million, of which IBRD US\$5.6 million, AIIB US\$5.6 million, GoS US\$2.8 million).** This component will finance capacity building and reform support activities to enable KWSC's evolution into a well-functioning utility that is resilient to climate change. The component will finance: (i) technical assistance and capacity building related to technical and commercial operations, including asset management, geographic information systems (GIS) and data systems, hydraulic modeling, NRW reduction, billing and revenue management, financial management, communications, and customer service; (ii) implementation of organizational structure reforms to enhance efficiency, accountability and commercial orientation in service delivery; (iii) implementation of a human resource reform plan, including a gender-inclusive recruitment and training strategy; (iv) preparation and implementation of water safety plans, including the establishment of a centralized water quality testing laboratory capable of carrying out biological and key chemical tests (including lead and other heavy metals); (v) rehabilitation and upgrade of office spaces and establishing an innovation, research, development, and learning center; (vi) technical assistance on PPPs, their outsourcing and management; (vii) preparation of all required rules, regulations and guidelines to enable full operationalization of the KWSC Act and their adoption as standard operating procedures for operationalizing the utility; and (viii) any other capacity-building activities necessary to strengthen KWSC's corporate governance and increase its operational and financial performance and adaptive capacity, such as technical trainings for women in the KWSC, support to the newly established Environmental and Social (E&S) cell at KWSC, and support to improve groundwater management.

33. **Component 2 –Safely Managed Water Supply and Sanitation Services (US\$958.56 million, of which IBRD US\$223.824 million, AIIB US\$223.824 million, GoS US\$241.912 million, private capital US\$269 million).** The component will finance investments to enhance Karachi's water supply and sanitation infrastructure. The medium-term investment plan addresses five interlinked structural problems in Karachi's water and sanitation system: a bulk water supply shortfall, a leaky water distribution network, an intermittent supply that does not meet drinking water quality standards, a lack of wastewater collection and treatment capacity, and persistent poor service levels, especially in Katchi Abadis.

¹⁶ The front-end fees (US\$600,000 each) of the World Bank and AIIB loans will be paid from the proceeds of each loan and are included in the total project cost.



34. **Sub-component 2.1: Provision of Safely Managed Drinking Water and Sanitation Services in Selected Unplanned Settlements (Katchi Abadis) (US\$28.88 million, of which IBRD US\$11.55 million, AIIB US\$11.55 million, GoS US\$5.78 million).** This sub-component will finance the expansion of safe drinking water and sanitation services in at least seven *Katchi Abadis*, benefiting over half a million people. Investments will include rehabilitation and expansion of water distribution systems, installation of ILC units, metered house connections, sewerage network rehabilitation and expansion, and accompanying social marketing and WASH behavior change communication. The KWSC will deliver services in partnership with local non-governmental organizations and community-based entrepreneurs as well as local women's organizations and transgender people's organizations.
35. **Sub-component 2.2: Priority Sewer Network Rehabilitation and Expansion (US\$23.22 million, of which IBRD US\$9.29 million, AIIB US\$9.29 million, GoS US\$4.64 million).** This sub-component will finance priority sewerage network rehabilitation and expansion to restore network integrity in critical areas and reduce sewer overflows and flooding that currently impact an estimated 3.02 million people in these areas. The networks for rehabilitation have been identified based on levels of deterioration, under-sizing, and combined sewer/ open channel sewers. These sewers carry wastewater to TP-I and TP-III, which are under rehabilitation by the KWSC.
36. **Sub-component 2.3: Priority Water Distribution Network Rehabilitation and Expansion (US\$56.93 million, of which IBRD US\$22.77 million, AIIB US\$22.77 million, GoS US\$11.39 million).** This sub-component will finance rehabilitation and upgrading of the water distribution network, including the creation of pressure zone boundaries and District Metered Areas for physical water loss monitoring, installation of ILC units, and installation of modern, digital network monitoring systems, including modern flow meters (district, domestic and bulk customer meters), pressure loggers, automatic metering reading and advanced metering infrastructure, and remote monitoring and sensing tools to monitor real-time residual chlorine levels. The sub-component will also finance network expansion to areas of Karachi that are currently not covered. In addition, the sub-component will finance a performance-based NRW reduction program to reduce physical and commercial water losses, prioritizing physical water losses by improving system maintenance, active leakage control, speed and quality of leak repairs, and pressure management. Reducing physical water losses will help reduce pressure on water sources and build resilience to droughts. An initial set of areas for rehabilitation has been established based on the potential for NRW reduction and energy-saving benefits relative to E&S complexity and costs. District Metered Areas financed under this sub-component, complemented by the installation of customer meters from the KWSC metering program, will support the implementation of NRW reduction plan.
37. **Sub-component 2.4: Improvement of Energy Efficiency (US\$19.90 million, of which IBRD US\$7.96 million, AIIB US\$7.96 million, GoS US\$3.98 million).** This sub-component will finance a range of energy-saving interventions to reduce energy consumption in KWSC's pumping stations and water treatment plants as well as the utilization of feasible renewable energy resources, and thus reduce energy costs and KWSC's carbon footprint. The KWSC will partner with competitively selected Energy Service Companies (ESCOs) to develop, design, build, and co-finance interventions that save energy, reduce energy costs, and decrease operations and maintenance costs at Dhabeji water pumping station, secondary pumps, and other major facilities. ESCOs will function as project developers for comprehensive energy conservation measures and assume the technical and performance risks associated with the selected interventions. ESCOs will be engaged on performance-based contracts with payments linked to actual energy cost savings. If needed, about US\$6 million is allocated for viability gap funding (VGF).
38. **Sub-component 2.5: K-IV Augmentation (US\$256.87 million, of which IBRD US\$77.75 million, AIIB US\$127.75 million, GoS US\$51.37 million).** This sub-component will finance: (i) 2.7 km of the



transmission line from reservoir 2 running from National Institute of Public Administration to the Expo Centre in Hasan Square in Karachi constructed as part of a common corridor; and (ii) apart from those 2.7 km, the construction of three transmission mains to bring 260 MGD of treated water into the existing network from three separate treated water reservoirs currently under construction as part of the ongoing government-financed K-IV Phase 1.

39. **Sub-component 2.6: Rehabilitation and Reconstruction of Existing Water Treatment Plants (US\$120.75 million, of which IBRD US\$29.50 million, AIIB US\$29.50 million, GoS US\$14.75 million, Private Sector US\$47 million).** This sub-component will: (i) finance the rehabilitation and upgrade of two water treatment plants (NEK-KII, 100 MGD; and Pipri JBIC, 50 MGD) with a total installed capacity of 150 MGD to ensure treated water entering the network from these plants is safe and meets national drinking water quality standard; and (ii) support the financing of the reconstruction of up to two water treatment plants (COD, 180 MGD, and NEK-KIII 100 MGD) with a total capacity of 280 MGD to increase treated water production design capacity. The KWSC intends to pilot the reconstructions through PPPs, lessons from which might enable future PPPs in other water treatment plants. A feasibility study indicates a total cost of US\$87 million for these two water treatment plant reconstructions. A preliminary assessment suggests an opportunity for a PPP supported by a VGF of US\$40 million. The VGF would finance payments against construction milestones outlined in the PPP contract. Of the estimated US\$47 million of private capital mobilized, which will comprise US\$14 million in equity and US\$33 million as debt, 84 percent (US\$40 million) is attributed to World Bank financing.¹⁷

40. **Sub-component 2.7: Construction of Malir Basin Wastewater Interceptor, TP-IV Wastewater Treatment, Recycling Plant, and new Recycled Water Distribution Network (US\$452 million, of which IBRD US\$65 million, AIIB US\$15 million, GoS US\$150 million¹⁸, private sector US\$222 million).** This sub-component will finance: (i) a wastewater interceptor (21.7 km) along the Malir River to capture the untreated wastewater currently discharged in the river and convey it to the TP-IV; (ii) TP-IV to treat 120 MGD influent of mixed domestic and pre-treated industrial wastewater compatible with a domestic wastewater treatment plant; (iii) a recycling facility with advanced tertiary treatment to produce 50 MGD water for industrial use; and (iv) a distribution network to convey recycled water to industrial users. These investments will improve the KWSC's ability to collect and treat wastewater in the Malir basin, reducing the pollution of the Malir River, which discharges into the Arabian Sea, and enhancing the resilience of the city's water supply system to droughts with recycled water. The KWSC intends to implement the full scope of this subcomponent as a single hybrid PPP contract and this subcomponent will contribute towards the US\$230 million of combined VGF from the IBRD, AIIB, and GoS (with financing from ADB under discussion). The ADB is advising the GoS on the PPP transaction with the issuance of a request for proposals expected in March 2025 and financial close estimated by March 2026. The VGF would finance payments against construction milestones outlined in the PPP contract. Of the US\$222 million private capital mobilized, which will comprise US\$67 million in equity and US\$155 million as debt, 50 percent (US\$111 million) is attributed to World Bank financing.¹⁹

¹⁷ This calculation is in accordance with the MDB Methodology for Private Investment Mobilization - Reference Guide - June 2018. The World Bank contributes US\$27 million or 84 percent to the total VGF from the two MDBs (WB and AIIB) of US\$32 million (AIIB US\$5 million or 16 percent). The GoS counterpart share is US\$8 million. It is assumed that 30 percent of private capital will come in the form of equity and 70 percent in the form of debt.

¹⁸ This amount includes US\$20 million of project counterpart financing under KWSSIP-2 as well as an additional US\$80 million of financing approved by the GoS cabinet on August 9, 2023 as part of the ongoing provincial S-III project (the revised PC-1 for S-III, which includes TP-IV, was approved on February 7, 2018) as well as an additional \$50 million of VGF under discussion with ADB.

¹⁹ This calculation is in accordance with the MDB Methodology for Private Investment Mobilization - Reference Guide - June



41. **Component 3 – Project Management and Studies (US\$25.24 million, of which IBRD US\$10.096 million, AIIB US\$10.096 million, GoS US\$5.048 million).** This component finances: (i) project management and coordination, including design review and construction supervision, project management support, E&S risk management, and project results monitoring and evaluation as well as beneficiary surveys and the Project's impact evaluation; and (ii) technical assistance for advance studies for potential subsequent water and wastewater infrastructure projects, including feasibility studies, engineering designs, E&S studies, and special studies on applicability of trenchless technology in water and wastewater infrastructure development in Karachi.

42. **Component 4 – Contingent Emergency Response Component (CERC) (US\$0 million).** The Borrower may request the Bank to reallocate project funds for the provision of immediate response to an Eligible Crisis or Emergency, as needed. This component would draw from the uncommitted loan resources under the project from other project components to cover emergency response.

C. Project Beneficiaries

43. **The MPA will provide an estimated 33.5 million people with all three aspects of safely managed WASH services (PrDO indicator).** Phase 1 targets 6 million people, Phase 2 targets 11.5 million people, and Phase 3 targets 16 million people. Under Phase 1, 16 million people in Karachi will receive safely managed water (PDO indicator i), and 7.5 million people will receive safely managed sanitation (PDO indicator ii). Phase 1 will also support hygiene behavior change communication for all citizens of Karachi. About half of Karachi's inhabitants are women, and 18 percent are youth (aged 15–24).

D. Rationale for Bank Involvement and Role of Partners

44. **The WBG brings a combination of financial, knowledge, and convening services that are critical to the success of the Pakistan Urban WASH Services MPA program.** The Bank has been a longstanding and valued partner in the Pakistan water sector for over 30 years. Through analytics and sustained engagement, the Bank has helped bring the issue of WASH service gaps and the link to child stunting to the forefront of policy discourse and helped catalyze and support a new generation of impactful WASH programs, such as KWSSIP-1 and achieving geographic convergency with other engagements (health, nutrition, etc.) that are relevant to reducing child stunting. The International Finance Corporation (IFC) is also active in the Pakistan water sector, with two advisory/market facilitation mandates for bulk water PPPs in Karachi²⁰ and promising leads on wastewater PPPs in Punjab, wastewater reuse for industrial parks in Karachi, and bulk water PPPs for Islamabad and Rawalpindi. The Multilateral Investment Guarantee Agency is exploring opportunities for political risk insurance products to support water PPPs in Pakistan.

45. **The World Bank plays a catalytic role in mobilizing additional concessional financing from other MDBs and the private sector.** The AIIB has committed to finance KWSSIP-2 with potential parallel financing from ADB. Investments will also help mobilize private capital as part of KWSSIP-2. Bank support to the MPA also includes private capital-enabling activities such as utility reform, capacity building, and efficiency improvements to enhance creditworthiness. KWSSIP-2 is expected to lay the groundwork to enable the mobilization of an estimated US\$225 million of private capital during the implementation KWSSIP-3, assuming 50 percent private financing of the US\$50 million required for the rehabilitation of two water treatment plants (Pipri Old, 105 MGD, and Gharo, 30 MGD) and of the US\$400 million

2018. The World Bank contributes US\$65 million or 50 percent to the total potential VGF from the three MDBs of US\$130 million (AIIB US\$15 million or 12 percent, ADB up to US\$50 million or 38 percent). It is assumed that 30 percent of private capital will come in the form of equity and 70 percent in the form of debt.

²⁰ Hub canal PPP (30 MGD) and Karachi District South Water Supply (65 MGD). Recently, the GoS dropped the hub canal PPP at bid stage and instead opted to proceed with public procurement.



estimated for Karachi District South Water Supply project currently being advised by IFC. Of the US\$225 million mobilized, 50 percent (US\$113 million) would be attributed to World Bank financing.²¹

E. Lessons Learned

46. **The design of the MPA draws on lessons and evidence from analytical work and more than five decades of WBG experience supporting WASH operations worldwide.** The project draws on a wealth of evidence that shows that lack of safe drinking water and sanitation and fecal contamination of the environment are the primary factors underlying the stunting crisis in Pakistan.²² The project scope includes investments to reduce the risk of fecal-oral transmission of pathogens through the provision of safe drinking water, safe wastewater collection and treatment, and hygiene behavior change.

47. **Lessons from global experience show that the World Bank is more impactful when it supports credible government programs that include infrastructure investments and institutional capacity strengthening.** This MPA shifts the default approach from ad hoc, opportunistic, and short-term engagements typically emphasizing infrastructure to a long-term programmatic approach combining critical infrastructure investments with reforms and capacity strengthening of service providers to maximize chances of achieving and sustaining safely managed services.

48. **Public utility reform initiatives in a complex socio-political context must deliver quick wins to maintain the support of all stakeholders.** The scope of Phase 1 includes short-term investments, such as ILC units, to enable the KWSC to immediately start providing safe drinking water in selected parts of its service area, support for increasing revenues rapidly and reducing costs through energy efficiency interventions and leveraging technology in billing and revenue collection.

49. **Finally, institutional reform requires sustained commitment from borrowers over the long term.** The project uses legal covenants to maintain focus and attention on the most critical reforms relevant to improving the KWSC's operational and financial performance. The project scope also includes complementary technical assistance and change management support to achieve results.

III. IMPLEMENTATION ARRANGEMENTS

A. Institutional and Implementation Arrangements

50. **Participating provinces and their corresponding PIUs will organize country-level annual learning events to discuss achievements, success, lessons learned and challenges in implementing the MPA.** This will be carried out through the country WASH knowledge platform under the MPA, in collaboration with the Bank. The final scope of the platform will be finalized in the first year of the project implementation.

51. **Phase 1 will retain the same implementation arrangements as KWSSIP-1.** The existing KWSSIP-1 Project Steering Committee (PSC), which is chaired by the Chairman of the Planning and Development Department of GoS and includes high-level representatives of all agencies involved in the project, will continue to provide strategic oversight and high-level coordination for KWSSIP-2. The KWSC will remain the secretariat to the PSC and will invite PIU directors to the PSC meetings, as needed, to update on the overall project implementation progress, including its business plans and reform activities. The KWSC Board of Directors will continue to champion and direct KWSC reform through an internal change management task force comprising senior KWSC managers. A performance agreement to be signed

²¹ Assuming the World Bank contributes 50 percent to the total VGF from the MDBs.

²² Mansuri, Ghazala, Mohammed Farhanullah Sami, Muhammed Ali, Hang Thi Thu Doan, Bilal Javed, and Priyanka Pandey. 2018. "When Water Becomes a Hazard: A Diagnostic Report on the State of Water Supply, Sanitation and Poverty in Pakistan and its Impact on Child Stunting." WASH Poverty Diagnostic Series Report, World Bank Group, Washington, DC.



between the GoS and KWSC will commit the KWSC (represented by its Board) to work towards the achievement of performance targets, including but not limited to the KWSSIP-2 results framework and updated targets from the 2019 CoC.

52. **The KWSC has not had significant capital investment for over a decade and will need to develop its capacity to manage large projects.** The KWSSIP-1 PIU has suffered from high staff turnover and struggles with M&E and project management. In addition to improved hiring, training, and management techniques and systems, the PIU will develop an updated Project Operations Manual including lessons learned from KWSSIP-1 prior to effectiveness. The KWSC will strengthen its PIU with additional technical and project management staff recruited from the market. In addition, the PIU will engage a Program Management Consultancy (PMC) firm with experience in managing complex capital programs to provide support in program coordination and managing interdependency risks.

53. **Existing PIUs managing the Karachi WSS capital program will remain but with more robust coordination arrangements.** The PIUs include the Water and Power Development Authority (WAPDA) PIU for the K-IV Phase 1, a Sindh Irrigation Department PIU for the Kalri Baghar Feeder Upper (KBFU) rehabilitation, and the KWSSIP PIU. A K-IV Coordination and E&S Implementation Committee, comprising the managers of the four major projects (KBFU, K-IV Phase 1, K-IV Augmentation Phase 1, and power transmission of 50 megawatts), shall be notified by the GoS to monitor and report on Environmental and Social Commitment Plan (ESCP) implementation and on construction progress on a quarterly basis. The GoS PPP unit and its transaction advisors will continue assisting the KWSC with the procurement and execution of all PPP projects. Further details of the implementation arrangements are in Annex 1.

B. Results Monitoring and Evaluation Arrangements

54. **Participating MPA provinces will be responsible for M&E through their respective PIUs.** The results frameworks will form the basis for tracking progress toward the PrDO. Existing M&E systems at each PIU will be strengthened to create a comprehensive Management Information Systems (MIS). All assets created will be geo-tagged. The MIS will include a data management system to process all information related to beneficiaries, disbursements, verification, and monitoring. Field staff, consultants, and beneficiaries will be able to directly upload information to the MIS in real-time through their smartphones, which PIU staff will then verify and consolidate centrally. Data will be aggregated across phases through the program-level MIS. Each phase of the MPA will finance the necessary equipment and tools, technical assistance, and capacity building to strengthen results and process monitoring at the project level, equip the PIUs to fulfill these responsibilities, and prepare regular progress reports.

55. **The World Bank, other interested or supporting development partners, and participating provinces will collaboratively conduct periodic program reviews to assess progress and course correction.** These regular reviews will also allow for a reassessment of government priorities to ensure continued support for the MPA and adjust ambition and coverage as needed.

C. Sustainability

56. **The program design supports sustainable outcomes.** First, participation is on a self-selection basis. Only those provinces that commit to the PrDO and to building credible service providers with the technical and financial capacity to sustain these services will join the program. Second, institutional capacity strengthening is included as a separate program pillar to draw attention to its importance as a critical factor in enhancing sustainability. Third, sustainability will require continuous support from provincial political leaders and civil service administrators, who may change over the life of the program. The World Bank will support the country WASH knowledge platform where leaders can share experiences



and benefit from change management support through the program activities. Finally, the project design has been informed by numerous studies to ensure that activities are of good technical design, least-cost, maximizing social and economic benefits, and are environmentally sustainable and climate resilient.

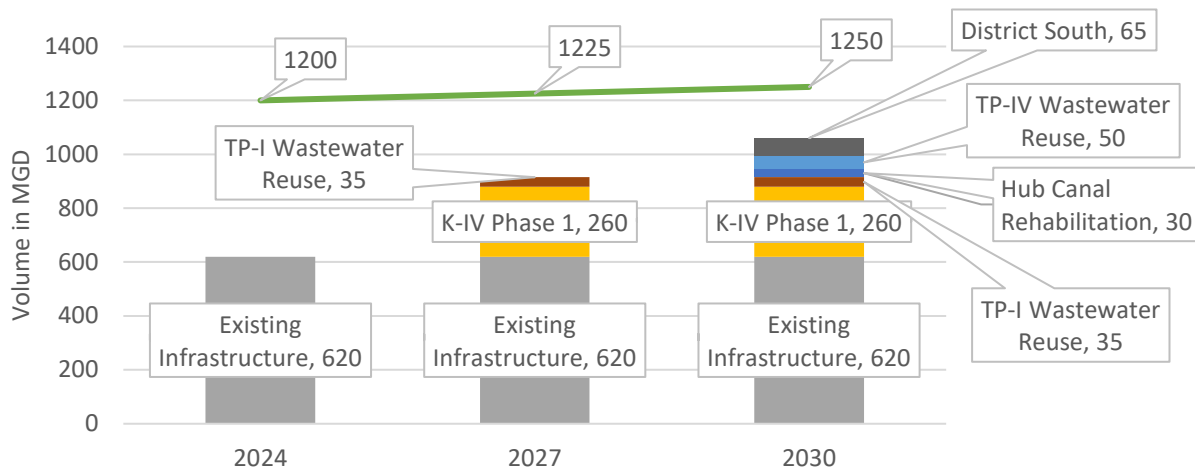
IV. PROJECT APPRAISAL SUMMARY

A. Technical, Economic, and Financial Analysis

Technical Analysis

57. The project is technically sound and adopts conventional WSS technology, incorporating resilience design principles and addressing critical infrastructure and institutional capacity gaps to improve service delivery.²³ The project will reduce the risk of fecal-oral transmission of pathogens through the provision of safe drinking water, safe collection and treatment of wastewater, and hygiene behavior change. Engineering designs and bidding documents are ready for around 60 percent of the project's investments, and KWSC has initiated procurement of the design review consultant for K-IV Augmentation (sub-component 2.5). The Bank has reviewed the cost estimates of all investments and found them reasonable. The ongoing institutional reform studies under KWSSIP-1, covering topics such as financial and human resources management, provide a basis for deepening reforms in KWSSIP-2.

Figure 1: KWSC Demand (line) and Supply (bars) Projections through 2030



58. The K-IV Phase 1 remains the most viable path to quickly reducing the water deficit in Karachi. K-IV Phase 1 is being financed by the GoP and the GoS (the estimated cost is PKR 126 billion or US\$410 million) and implemented by WAPDA, who will also operate it for the first year. K-IV Phase 1 will withdraw an additional 260 MGD from Keenjhar Lake and convey the water for approximately 110 km via pipes to three new water treatment plants and reservoirs on the outskirts of Karachi. Keenjhar Lake is a Ramsar site that already serves as Karachi's main water source and is supplied primarily by water from the Indus River from Kotri Barrage via the KBFU Canal. The K-IV Phase 1 requires 50 megawatts of power to be supplied via a 30 km transmission line and a grid substation to be designed and constructed by the Sindh Transmission and Dispatch Company (STDC). The STDC has hired technical consultants and an E&S consulting firm. The proposed sub-component 2.5 works will connect the K-IV Phase 1 ground storage reservoirs to Karachi's existing water distribution network. It is expected that the K-IV Phase 1, including

²³ Engle, Nathan Lee, Medina Daniel, Greg Felter, and Nelson Sean. 2020. "Resilient Water Infrastructure Design Brief." Water Global Practice Working Paper. World Bank Group, Washington, DC.



its power supply, would be completed by December 2026. Additionally, the GoS is pursuing the rehabilitation of an existing bulk water supply canal (Hub Canal), which would increase supply by 30 MGD. Figure 1 summarizes the KWSC's planned investments to close the bulk water supply deficit.

59. **However, given the competing needs from agricultural water use for water in Keenjhar Lake, there is a risk that the K-IV Phase 1 will not always be capable of supplying the additional bulk water to Karachi at the required urban water supply reliability level (95 percent).** To mitigate this risk, the Sindh Irrigation Department has recently started the rehabilitation of the KBFU canal to restore its hydraulic capacity, which is expected to increase water delivery to Keenjhar Lake by at least 260 MGD, enabling reliable supply to K-IV Phase 1 while maintaining existing agricultural water withdrawals and safeguarding the lake's ecology. The project will cost an estimated PKR 40 billion (US\$142 million) and is being financed equally by the GoS and GoP. The KBFU rehabilitation works expected completion date is June 2027, which aligns with the expected completion date of the K-IV Phase 1 and K-IV Augmentation.

60. **The KWSC has programmed the distribution network rehabilitation investments to ensure that critical network upgrades are complete before K-IV Augmentation comes online.** In addition to targeting old, damaged, and obviously leaking network segments, investments will be informed by a calibrated network hydraulic model supported by the project to analyze how the network may be modified to serve currently unserved areas to control water losses and save energy. The model will be used to simulate the benefits of the interventions in terms of reduced losses, increased hours of service, and pressure control.

61. **Rehabilitating water treatment plants will ensure that treated water entering the network meets the national drinking water quality standards.** In addition, the project will deploy ILC units at scale in the network to disinfect water before the collection point. ILCs are a cost-effective way to address microbial water quality risks in the network as part of a comprehensive water safety plan. Evidence from randomized control trials suggests that ILC units can reduce child diarrhea by up to 25 percent.²⁴

62. **The operation is aligned with the mitigation and adaptation goals of the Paris Agreement.** Regarding *mitigation*, all project activities are on the Universally Aligned List. The electricity utilized for the WASH services will be sourced from the grid and not utilize on-site fossil fuels as the main energy source. Regarding *adaptation*, Karachi is extremely vulnerable to water scarcity, extreme heat, flooding, sea level rise, and earthquakes. The program and the project intend to address climate risks in the project area to limit their impacts on the infrastructure and beneficiaries. The risks to WSS infrastructure will be reduced to an acceptable level by integrating the resilient design principles²⁵ within project design, ensuring that climate change and disaster risks are adequately considered in an interactive and participatory manner.²⁶ Such design principles include the use of submersible pumps, flood-proofing hard components, water quality and quantity data collection, structural bracing and/or use of reinforcements as applicable to project component thus enhancing the reliability of the water supply and sanitation services provided, especially during extreme weather events. The project will decrease the risk of climate change-aggravated flooding and droughts through supporting climate-resilient water supply network

²⁴ Pickering, Amy J., Yoshika Crider, Sonia Sultana, Jenna Swarthout, Frederick G. Goddard, Syed Anjerul Islam, Shreyan Sen, Raga Ayyagari, and Luby, S. P. 2019. "Effect of In-line Drinking Water Chlorination at the Point of Collection on Child Diarrhea in Urban Bangladesh: A Double-blind, Cluster-randomized Controlled Trial." *The Lancet. Global health* 7 (9): e1247–e1256.

²⁵ This 2019 World Bank Resilient Water Infrastructure Design brief will be used to ensure the WSS infrastructure is itself resilient to floods, droughts, and high winds.

²⁶ Measures that may be considered include: reinforce surface water intakes, install flood barriers around the plant, installing pumps if needed to drain the area inside the barriers, elevating water tanks, pumping stations and electrical components, installing submersible pumps, developing procedures to adapt to source waters with higher pollutant loads, installing capacity to manage extreme changes in influent quality, etc.



rehabilitation (including NRW reduction) and expansion, duly considering current and projected climate change impacts, leak reduction, and efficient use of water sources. In addition, positive public health outcomes, including reduced child stunting, are boosted through the increased collection and treatment of wastewater and fecal sludge by the project. Lastly, the introduction of water reuse will increase available water, contributing to resilience to droughts. The CERC Manual provides the overall financing framework and types of activities to be financed under CERC. The task team will ensure that all eligible activities included in the CERC Manual/CERC Annex of the Project Operations Manual are Paris aligned.²⁷ Immediate emergency preparedness and response activities that are temporary and time-bound (such as debris removal, emergency evacuation, structural stabilization, restoration of essential services, procurement of medicines, food security, and cash transfers) are considered low risk and therefore Paris aligned. Thus, risks from climate hazards have been reduced to an acceptable level and are not likely to have material impact on the operation and its development objectives.

Economic Analysis

63. **The economic rationale for this project is strong, with a positive net present value (NPV) of US\$1,140 million and an economic internal rate of return (IRR) of 35 percent.** The benefits considered include reduced cost of using alternative water sources, reduced time spent collecting water, and health benefits from reducing waterborne diseases. However, the direct environmental benefits of reducing pollution cannot be estimated reliably. Sensitivity analyses showed that the project is robust and that even significant changes in critical variables will not render the project unviable. For instance, a one-year delay in completing the K-IV Augmentation subproject critical to water service improvements only reduces the economic IRR of the subproject from 22 percent to 17 percent.

64. **Greenhouse gas (GHG) ERs—monetized by applying a low and high shadow price of carbon—further add to the project's economic benefits.** The project is expected to generate a net ER of 17,558,840 tons of carbon dioxide (CO₂) equivalent over its economic lifetime. The annual GHG reduction is estimated at 877,942 tons of CO₂ equivalent. The ERs are driven by energy efficiency gains, reduced tanker water deliveries, and improved wastewater collection and treatment²⁸.

Financial Analysis

65. **The project is financially viable for KWSC with an NPV of US\$470 million and IRR of 20 percent, assuming no cost of financing to KWSC.** Although the KWSC Act enables the KWSC to operate on a commercial basis and to borrow to meet its capital expenditure and recover all its costs (including financing costs) through tariffs, the KWSC is still a long way from achieving full cost recovery. It will continue to depend on the GoS for capital financing for some time, especially for wastewater operations.

66. **The KWSC has not produced complete audited financial statements, so the World Bank could not reliably ascertain its financial position.** For the FY08-20 statements, the auditors have provided disclaimers due to insufficient audit evidence. An issue of concern is the ability of the KWSC to operate as a solvent and financially viable entity during and after the project, with reduced dependency on operational GoS subsidies. Operating cashflows, excluding subsidies, have remained negative for the past years. The KWSC faces the challenge of increasing revenue from its water sales due to an inaccurate and incomplete customer database, lack of customer metering, and archaic billing and collection practices.

²⁷ This would then be ensured for the specific activities to be financed and included in the Emergency Action Plan, based on the information available at the time of activation of CERC.

²⁸ A full analysis and estimate of the GHG were concluded from the use of the corporate model and further documented in the Climate Change Technical Note and project files.



Although billings increased significantly since 2018, collection efficiency has remained stagnant at around 70 percent. These trends point to opportunities for the KWSC to improve its financial performance and become a creditworthy entity capable of attracting capital, should it pursue more ambitious reforms to increase its revenues and control operating costs. With a large potential customer base in Karachi, the KWSC can improve its performance over time and recover the full costs of water supply services.

67. **The project will accelerate the KWSC's progress toward cost recovery through NRW reduction, improved billing and revenue collection, energy efficiency, and other reforms to optimize costs.** The World Bank estimates that these interventions will increase the KWSC's operating cost coverage ratio (OCCR)²⁹ for water supply from 75 to 150 percent (based on the outcome of a financial model incorporating projected improvements from the current baseline) by the end of the project. To maintain transparency and focused attention on the financial turnaround, the KWSC has agreed to: (i) prepare, within one year of project effectiveness, a financial sustainability roadmap based on a robust utility financial model with time-bound actions to achieve an OCCR for water supply of at least 130 percent (including electricity costs); and (ii) adopt a revised water and wastewater tariff framework consistent with the financial sustainability roadmap approved by the KWSC Board. Measures for increasing cost recovery may include financial restructuring, promotion of private sector participation, promotion of efficiency, and adjustments of the structure or levels of tariffs. The project will support advisory services to evaluate options for achieving financial sustainability as part of the scope of the performance audit.

B. Fiduciary

68. **The project financial management (FM) arrangements meet the requirements of the World Bank Policy for IPF, and risks are moderate.** The FM unit under the existing PIU is sufficiently staffed and is headed by an FM Specialist. An assessment of the KWSC's FM arrangements concluded that it maintains FM arrangements that are acceptable to the World Bank and that, as part of the overall arrangements in place for implementing the project, it provides reasonable assurance that proceeds of the loan will be used for the purposes for which they are allocated. The assessment provides reasonable assurance that finances are used for the intended purposes, with due attention to the principles of economy, efficiency, effectiveness, transparency, and accountability. The KWSC has a detailed chart of accounts for recording transactions that will be used for the project accounting. A unified public FM system exists at all levels of government in Pakistan. Funds from the World Bank will be transferred to the project's designated account (DA), established per the Revolving Fund Assignment Account Procedure, and released through the Finance Department to the Implementing Agency. Annual external FM audits will be conducted by the Department of Auditor General and submitted to the World Bank by December 31 each year.

69. **The project will follow traditional transaction-based IPF with a total financing of US\$240 million from IBRD and US\$240 million from the AIIB.** The project will benefit from FM software already implemented in KWSSIP-1, which can generate financial reports directly from the system in the agreed-upon format, i.e., Interim Unaudited Financial Reports (IUFRs). The project will submit the IUFRs on a semester basis within 45 days from the end of each calendar semester. Disbursements can be in direct payments for foreign currency payment outside Pakistan and report-based, with advances equivalent to six months' forecasted expenditure. This disbursement will be processed based on the provision of six monthly IUFRs containing cash forecasts. The World Bank will conduct FM reviews regularly to ensure that

²⁹ The OCCR is defined as the ratio expressed in percentage of collected revenues to the total operating cost but not including debt service or depreciation cost. The projected OCCR is indicative and will be confirmed based on the audited financial statements that meet international standards, expected at the end of year 1. The OCCR is expected to double from the baseline assumption at the end of the project as a result of performance improvements and envisaged tariff adjustment.



expenditures incurred are eligible.

70. **Retroactive Financing.** All project activities except subcomponents 2.5 and 2.7 are eligible for up to US\$43.5 million of retroactive financing. Activities under subcomponent 2.5(i) are eligible for up to US\$4.5 million of retroactive financing. Retroactive financing for sub-component 2.5(i) would be subject to the completion of an E&S audit and corrective measures for works completed up to the request for withdrawal as per the ESCP.

71. **The project, including VGF components, will follow the World Bank's Procurement Regulations for IPF Borrowers** (Procurement in Investment Project Financing, Goods, Works, Non-Consulting and Consulting Services, Fifth Edition, September 2023). Some procurement activities may follow the national procurement procedures (national competition) subject to conditions specified in the textual part of the procurement plan approved by the World Bank. The project will be subject to the World Bank's Guidelines on Preventing and Combating Fraud and Corruption in Projects Financed by IBRD Loans and IDA Credits and Grants (revised as of July 1, 2016, Anti-Corruption Guidelines). The updated procurement risk assessment of the KWSC is detailed in Annex 1, including risk areas and mitigation measures agreed.

72. **The KWSC has prepared a draft 18-month procurement plan and a Project Procurement Strategy for Development (PPSD)** that addresses the overall procurement outlays, market availability, system constraints, procurement competencies, and implementation aspects of the project. Identified procurements include: (i) large infrastructure works for K-IV Augmentation (four lots costing US\$68 million, US\$99.19 million, US\$32.87 million, and US\$56.81 million); (ii) contracts for other works using national and international bidding process (US\$4–30 million); (iii) design and supervision consultancies costing US\$4–6 million; (iv) other consultancies including energy audit follows up, communication strategy, M&E, and project management consultancy costing US\$1–3 million; and (v) minor goods. The project will also finance VGF for PPPs under development. According to the Private Participation in Infrastructure database, Pakistan has closed 87 PPP projects in the last 20 years, but none in the water sector. Sindh province has awarded nine PPP projects in the transport, power, and health sectors. Market feedback from recent PPP projects tendered by the GoS is that the lack of an international arbitration clause in the contract rendered the project unattractive to international and local players. Currently, the Sindh cabinet is in the process of reforming statutory limits on international arbitration. Lessons learned from the ongoing TP-I tendering will inform TP-IV. The World Bank will conduct due diligence to confirm the technical and commercial aspects of the PPP, including the rationale and size of the VGF, affordability of tariffs, and market acceptability, and the PPCSD will be updated accordingly.

C. Legal Operational Policies

Legal Operational Policies	Triggered?
Projects on International Waterways OP 7.50	Yes
Projects in Disputed Area OP 7.60	No

73. **OP 7.50 (Projects on International Waterways) is triggered because the project involves the use of water from the Indus River, which is considered an international waterway.** The project activities include rehabilitation and expansion of the existing water distribution network in Karachi and its Katchi Abadis, bulk water augmentation, and rehabilitation and upgrade of the existing water treatment plants.



These investments do not change the nature of the existing schemes and will not adversely change the quantity or quality of water flows to other riparians. Moreover, the terms of reference for the studies and assessments on water infrastructure and groundwater and the preparation of KWSSIP-3 will include an examination of any potential riparian issues. Accordingly, an exception to the notification requirement under Paragraphs 7 (a) and (b) of the Policy was approved on August 8, 2024.

D. Environmental and Social

74. **The World Bank has assessed the project's potential E&S risks and impacts.** The project will provide safely managed WASH services and reduce environmental pollution from untreated wastewater. For the PPP activities in component 2, OP 4.03 (Performance Standards for Private Sector Activities) will be applied. The overall E&S risk is High due to social risks related to legacy resettlement issues in the associated facility and potential economic resettlement and livelihood impacts on non-titled people on public land and right of way during project implementation.

75. **The project's environmental risk is Substantial.** Risks and impacts during construction include: (i) an increase in air and noise pollution; (ii) loss of trees and vegetation; (iii) generation of construction wastes and wastewater; (iv) quarrying activities and associated soil erosion; and (v) occupational health and safety (OHS) and community health and safety (CHS) issues, including traffic disruptions due to the activities in densely populated areas. During project operation, risks and impacts include the generation of sludge from wastewater and water treatment, OHS hazards related to chemicals management, potential soil, groundwater, and water distribution contamination due to any leakages of the sewerage pipes, and over-abstraction from Keenjhar Lake without offsetting or regulating additional withdrawals for K-IV Phase 1. Additionally, components of the project may also be exposed to flash floods, which will be considered while finalizing the project designs.

76. **The project design includes actions to mitigate potential adverse environmental impacts on the Keenjhar Lake due to increased withdrawals.** A Keenjhar Lake Water Balance Study and an Ecological Assessment for Keenjhar Lake and Downstream Indus Impacts confirmed that the most significant mitigation action for Keenjhar Lake is the rehabilitation of KBFU, which is ongoing. As per the ESCP, the Keenjhar Lake water balance will be regularly updated, and increased water delivery to the lake will be realized before an equal or lesser amount is withdrawn for K-IV Phase 1. The ESCP also requires implementation of the Keenjhar Lake Conservation Action Plan to begin before award the civil works under subcomponent 2.5(ii) and to be maintained throughout project implementation. The K-IV Coordination and E&S Implementation Committee will coordinate and report on ESCP compliance. The Ecological Assessment found that the increases in withdrawals from the Indus at Kotri Barrage to offset K-IV withdrawals at Keenjhar Lake represent no more than 3 percent of historical withdrawals at Kotri. The contribution of the project to the cumulative impacts of multiple withdrawals and degradation of the Indus watershed is therefore assessed as negligible. The planned irrigation improvements and improved water resources planning under the ongoing World Bank-financed Sindh Water and Agriculture Transformation Project (P167596) and Sindh Barrages Improvement Project (P131324, and its Additional Financing P179051) are expected to mitigate regional issues of overallocation and competition for water.

77. **The project's social risk is High, mainly due to legacy resettlement issues of the associated facility and potential livelihood, accessibility, and CHS issues, as the project will involve works in densely populated areas.** The World Bank, in close collaboration with the KWSC and the PIU, has confirmed that no Anti-Encroachment Drives resulting in forced evictions have taken place on the direct footprint of the project. In addition, the project does not anticipate private land acquisition as the required land and Rights of Ways, including of transmission lines, are already owned by the KWSC or the GoS. Additionally, the GoS



has recently adopted the 2022 Resettlement and Rehabilitation Policy, which is fully aligned with the World Bank's ESF and which further mitigates social risks. Civil works in the densely populated and congested urban environment of Katchi Abadis may result in temporary negative livelihood impacts due to civil works (e.g., due to restricted access) on businesses, transporters, vendors, hawkers, etc. Rehabilitation of the water supply and sewerage network also poses the issue of severance of informal connections. This will be mitigated through a communications campaign for a clear connection regularization procedure and enforcement of the law. The project investments also pose labor-related risks and impacts (e.g., working conditions, OHS, contract terms, etc.). While locally available labor will be preferred, an influx of labor may be anticipated and labor camps may need to be established. Additional social risks include the exclusion of vulnerable communities from project benefits and the lack of meaningful engagement and consultation with communities, particularly vulnerable groups such as women, the elderly, minorities, and the poor. The project will mitigate the social impacts of any increases in water tariffs first through improved service, supporting the KWSC's communications capacity, and by including affordability considerations in tariff setting.

78. Legacy Resettlement Issues and ESF Gaps for K-IV Phase 1 (Associated Facility): Based on the ESF, the government-financed K-IV Phase 1 is an Associated Facility. Even though it is not financed by the World Bank, it must comply with the World Bank's ESF. A detailed gap assessment between government requirements and actions and the World Bank's ESF requirements was concluded in December 2023 for the water conveyance system currently under construction. The most significant gap identified is related to the social legacy of low compensation rates for land acquisition. A Corrective Action Plan was developed, and the GoS has agreed to retroactively compensate project-affected people in accordance with the requirements under Environmental and Social Standards (ESS) 5. Other gaps identified require variation orders to the existing contracts between WAPDA and the K-IV Phase 1 contractors for remedial measures such as increasing E&S staffing, adopting labor management plans, and establishing a Grievance Redressal Mechanism (GRM). All corrective actions in the gap analysis must be completed before K-IV Augmentation works are awarded. The STDC will conduct an Environmental and Social Impact Assessment (ESIA) for the K-IV Phase 1 power transmission line in alignment with the World Bank ESF and develop a Land Acquisition and Resettlement Plan (LARP). Private land acquisition is not envisaged for the transmission line. All E&S instruments to be developed for the transmission line will be reviewed and endorsed by the World Bank; the ESIA will become part of the bidding documents for the engineering, procurement, and construction contracts; and payment of compensation under the LARP will be completed prior to the award of the K-IV Augmentation works under subcomponent 2.5(ii).

79. To address E&S risks, the KWSC has prepared an Environmental Management Framework and Social Management Framework that covers both KWSSIP-1 and KWSSIP-2, and additional E&S instruments for KWSSIP-2 in line with World Bank ESF requirements.³⁰ The documents identify mitigation measures for E&S risks to be incorporated in the subproject design and measures for managing and monitoring residual risks throughout the implementation of the project. These documents include:

- **A ESIA for the Malir Basin Interceptor and TP-IV Wastewater Treatment Plant** was disclosed locally on April 25, 2024, and on the World Bank website on May 13, 2024. The ESIA will be updated during project implementation once the design of the PPP is finalized.
- **A ESIA for K-IV Augmentation** was disclosed locally on April 25, 2024, and on the World Bank website on May 21, 2024. This ESIA is informed by the Keenjhar Lake Conservation Action Plan.
- **A Corrective Action Plan** to align the resettlement planning and implementation of the government-

³⁰ Local disclosures are on the KWSSIP website. ESF documents required to be translated into Urdu have been disclosed locally.



financed K-IV Phase 1 with the World Bank's ESF was disclosed locally on May 27, 2024, and on the World Bank website on June 27, 2024. Corrective Action Plan payments to private landowners will be completed before works on K-IV Augmentation construction begins.

- **A Labor Management Plan** was disclosed locally on May 30, 2024, and on the World Bank website on June 27, 2024.
- **A merged Resettlement Plan** covering Malir Basin wastewater interceptor, priority sewer network, and K-IV Augmentation works was disclosed locally on May 27, 2024, and on the World Bank website on June 27, 2024.
- **An Abbreviated Resettlement Plan for Katchi Abadis** was disclosed locally on July 24, 2024, and on the World Bank website on July 31, 2024.
- **A Stakeholder Engagement Plan** was disclosed locally on May 27, 2024, and on the World Bank website on June 27, 2024.
- **Five Environmental and Social Management Plans (ESMPs)** were disclosed locally as of July 20, 2024. The five ESMPs were for the following sub-components: Provision of Safely Managed Drinking Water and Sanitation Services in Selected Katchi Abadis; Priority Water Distribution Network Rehabilitation and Expansion; Priority Sewer Network Rehabilitation and Expansion; Improving Energy Efficiency; and Rehabilitation and Reconstruction of Water Treatment Plants.
- The **ESCP** was first disclosed locally on August 11, 2024, and updated on November 8, 2024. On the World Bank website, it was first disclosed on August 26, 2024, and updated on November 14, 2024.
- The **Appraisal Environmental and Social Review Summary** was disclosed on the World Bank website on August 26, 2024.

80. **Grievance Redress Mechanism: A GRM is already established under KWSSIP-1 and will be retained for KWSSIP-2 with minor modifications.** If required, modifications may be made to the existing GRM and web-based GRM portal to better address KWSSIP-2 activities. The GRM is designed to collect, collate, review, and redress stakeholders' concerns, complaints, and grievances. Dedicated communication materials will be developed to help stakeholders become familiar with the grievance redress channels and procedures. The GRM will be accessible and understandable for all project stakeholders for the entire project life. A labor-specific GRM will be operationalized by the contractor at each sub-project site, with contextualized grievance registration channels for the laborers/workers.

81. **The KWSSIP-2 PIU will include one Senior Environmental Specialist, one Senior Social Specialist, one Senior Gender Specialist, one OHS Specialist, and four field staff including two Environmental Specialists and two Social Specialists. In addition, the KWSC will continue to strengthen its E&S Cell.** The PIU E&S team is responsible for the implementation of ESMPs, resettlement plans, and other E&S instruments, and they will ensure the contractor's E&S requirements and bills of quantities are included in the contract documents, approve the contractor's ESMPs, and supervise their implementation. Comprehensive training in E&S instruments for new PIU and KWSC staff will be conducted during project implementation. Contractors must engage adequate environmental, social, health, and safety specialists for project implementation. All institutional arrangements for the KWSC E&S Cell will be in place within one year of project effectiveness, including an E&S Director, Environmental, Social, Health, and Safety Specialists, support staff, and required permanent office arrangements. The K-IV Coordination and E&S Implementation Committee members will implement ESCP actions outside the purview of the PIU.

82. **Gender-based violence (GBV) and sexual exploitation and abuse (SEA): The GBV/SEA risk is assessed as moderate.** Most construction activities will occur in congested urban areas where the risk of occurrence of GBV/SEA is reduced. However, some will take place in rural/remote areas where there is a comparatively higher GBV risk and poor support systems for survivors. Additional risks include potential



exclusion of and lack of meaningful engagement with vulnerable groups. The low capacity of the PIU to manage social issues also raises the risk. To mitigate these risks, all workers will be required to sign a Code of Conduct and will receive training on GBV and harassment awareness. A SEA-sensitive GRM will be staffed by trained personnel, and service providers will be mapped to ensure appropriate responses to survivors. Additionally, the surrounding community will be informed about the risks and the GRM. Hiring a senior gender specialist in the PIU and E&S training programs will address these gaps as well.

83. Citizen Engagement: The project explicitly seeks to support the engagement of targeted beneficiaries and other stakeholders through consultative processes and feedback mechanisms, which will help adjust the approach to their inclusion and maximize benefits to vulnerable individuals. During project preparation, various stakeholder groups were consulted for the preparation of the E&S documents through focus group discussions, interviews, and meetings. Separate meetings with women were arranged to obtain their views and concerns. In addition, stakeholder workshops were also arranged, a broad range of institutional and grassroots stakeholders were invited, and their views, suggestions, and concerns were obtained. Civil society organizations were also invited to these workshops, and dedicated sessions were also arranged with the key civil society organizations to obtain their views and to benefit from their experience, particularly relating to working in Katchi Abadis. Extensive community consultations have also been conducted while designing the project interventions in the Katchi Abadis. Similar citizen and stakeholder engagement will be continued throughout the various phases of project implementation, details of which are provided in the Stakeholder Engagement Plan. The project results framework includes a customer satisfaction indicator which will be assessed through independent customer surveys commissioned by KWSC.

84. Gender: The project will build on KWSSIP-1 efforts to increase the recruitment and promotion of women in technical positions in KWSC as per the CoC.³¹ The target for recruiting women in technical positions in KWSSIP-1 was delayed due to a hiring moratorium by the GoS since 2008 and delayed implementation of the Gender Action Plan (GAP) developed during KWSSIP-1. Achievements under the GAP include building women-specific bathrooms and prayer areas, on-site daycare support, and skills training for KWSC employees. With the lifting of the hiring moratorium and renewed momentum in the GAP, KWSSIP-2 will continue to address gender gaps in employment, representation, and leadership at the KWSC. Under Component 1, the project will continue to focus on recruiting more women in technical and decision-making positions by developing a gender-inclusive recruitment strategy as part of human resource reforms at KWSC. KWSSIP-2 will support technical training explicitly targeting women in the KWSC to help them progress to higher-level positions. The project will also support institutionalizing an internship program for women graduates from technical universities, creating pathways for employment at KWSC. To foster a safe and equal work environment, KWSSIP-2 will ensure that all employees go through training on SEA/sexual harassment. The project will establish a transparent complaint process and a trained inquiry committee. KWSSIP-2 will track the number of newly recruited staff in technical and managerial positions and support promoting women employees to higher-grade positions.

V. GRIEVANCE REDRESS SERVICES

85. Grievance Redress. Communities and individuals who believe that they are adversely affected by a project supported by the World Bank may submit complaints to existing project-level grievance mechanisms or the Bank's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project affected communities and

³¹ The CoC states that by 2030 at least 10 percent of employees in each grade must be female.



individuals may submit their complaint to the Bank's independent Accountability Mechanism (AM). The AM houses the Inspection Panel, which determines whether harm occurred, or could occur, as a result of Bank non-compliance with its policies and procedures, and the Dispute Resolution Service, which provides communities and borrowers with the opportunity to address complaints through dispute resolution. Complaints may be submitted to the AM at any time after concerns have been brought directly to the attention of Bank Management and after Management has been given an opportunity to respond. For information on how to submit complaints to the Bank's Grievance Redress Service (GRS), visit <http://www.worldbank.org/GRS>. For information on how to submit complaints to the Bank's Accountability Mechanism, visit <https://accountability.worldbank.org>.

VI. KEY RISKS

86. **The overall risk rating for the MPA is Substantial** considering the risks inherent in program activities in all phases, including network infrastructure investments in dense urban contexts. The Phase 2 E&S risk is rated substantial, while for phase 3 it is rated high. Macroeconomic risk is also rated High for phases 2 and 3. The Phase 2 and 3 risks on institutional capacity, fiduciary, and stakeholders are expected to reduce as a result of KWSSIP-2 risk mitigation measures. Moreover, the Program learning agenda will help strengthen measures for the MPA to reduce potential risks that would be revealed.

87. **The overall risk rating for KWSSIP-2 is Substantial due to the following substantial and high risks.**

88. **Political and Governance risk is rated High.** The project's political context is complex, with multiple public and private actors. The risk of political disagreements affecting implementation is high, especially on contentious topics such as project target areas, tariff revisions, and PPPs. To mitigate these risks, the preparation process has emphasized a collaborative approach between the GoS, KWSC, and Karachi Municipal Corporation. This approach will continue through the PSC, which brings diverse political actors together. While the extent to which project-specific measures can mitigate political risks is limited, the project includes specific actions to build public support through broader stakeholder participation, strategic communication, and outreach. Governance risks will be mitigated through: (i) implementation of the institutional strengthening activities under the project; (ii) dialogue and close collaboration with the GoS and KWSC, as well as proactive engagement by the World Bank team, and management as issues arise; (iii) use of legal covenants to maintain focus on critical reforms and the outcomes of the reforms; and (vii) leveraging digital technologies tools and geospatial information systems in M&E.

89. **Macroeconomic risk is rated High.** While Pakistan's economy has stabilized over the past 12 months due to new external inflows and improvements in macroeconomic management, debt and external financing needs remain elevated, and risks remain very high. Significant risks include international price or interest rate shocks; natural disasters that may impact domestic production, revenue collection, or expenditure needs; reversal of recent reforms to constrain expenditures or to ensure exchange rate flexibility; and lack of progress in further structural reforms in the fiscal, state-owned enterprise, and energy areas. Realization of risks may lead to macroeconomic instability, with major impacts on economic activities, prices, and household incomes, thereby impeding achievement of project results. Risks will be partially mitigated by ongoing World Bank and other partners' support to structural policy reforms in the fiscal, energy, and state-owned enterprise reform sectors. Mitigation measures also include the financing provided by the project through the IPF instrument and the monitoring by the PSC that resources will remain available during the implementation of the project. Risks will also be mitigated in the following ways: (i) project costing includes substantial price contingencies to mitigate the risk of an increase in the price of imported goods; and (ii) counterpart financing is limited to about one-quarter of the total project cost. However, the GoS has related obligations of approximately



US\$28 million for the K-IV Phase 1 (associated facility) and US\$71 million for the financing of the KBFU Rehabilitation, therefore, even with these aforementioned measures which can be further exacerbated by the inability of GoS to secure funds to meet its financial obligations, the residual risk remains high.

90. **Environment and Social risk is rated High.** The combined E&S risks are rated high due to social risks associated with legacy resettlement at the associated facility. A Corrective Action Plan has been developed to mitigate these risks. Additional social risks from the potential economic displacement and livelihood impacts due to works in densely populated neighborhoods will be mitigated by implementing the merged Resettlement Plan and Abbreviated Resettlement Plan for Katchi Abadis. Substantial environmental risks include the potential for ecological impacts, mainly on mangroves at the mouth of the Malir River and on Keenjhar Lake due to water extraction for K-IV Phase 1. This risk is being mitigated by the ongoing rehabilitation of KBFU canal and ESCP actions requiring regular water balance studies of Keenjhar Lake and implementation of the Keenjhar Lake Conservation Action Plan, as well as increased water delivery to Keenjhar Lake before an equal or lesser amount is withdrawn for K-IV Phase 1. Before contract award for civil works for K-IV Augmentation, adequate funds for the KBFU rehabilitation and K-IV Phase 1 works must be secured to avoid material delay and ensure their timely completion. Failure may lead holding disbursement to pursue financing the K-IV Augmentation works. Additionally, the risk to mangroves at the mouth of the Malir River will be mitigated by controlling access to the mangroves from the TP-IV site, ensuring that no mangrove cutting takes place.

91. **Institutional Capacity for Implementation and Sustainability risk is rated Substantial.** The KWSC suffers from high staff turnover and struggles with M&E and management of project interdependencies. These risks will be mitigated by: (i) hiring additional technical project management staff from the market; (ii) training PIU management and staff; (iii) improving PIU management and administrative systems; and (iv) engaging a PMC to provide support in managing coordination and project interdependencies.

92. **Fiduciary risk is rated Substantial.** Procurement risk is rated as substantial due to the size of the K-IV Augmentation works contracts and the PPP interventions. The procurement risk will be mitigated through proper staffing of the PIU with procurement and contract management specialists, training and capacity building in procurement and contract management, and engagement of a PMC which will, inter alia, support the PIU in procurement and contract management.

93. **Stakeholders risk is rated Substantial.** The sector includes many stakeholders whose interests are often difficult to reconcile. A wide array of formal and informal organizations, private companies, non-governmental organizations, unions, religious institutions, and local government structures can be expected to vie for influence on project decisions and contracts. The project is financing a sustained, professional communications and outreach effort. The selection of Katchi Abadis could lead to controversy and opposition from local interest groups (e.g., current water vendors) and delay implementation. The KWSC will work with local councilors and engage with residents through the KWSC's Katchi Abadi Cell. The KWSC will also establish a small local office in each of the selected Katchi Abadis and a special Katchi Abadi counter in its service centers.

94. **Coordination risks are rated Substantial due to interdependencies that may cause delays and cost overruns.** The interdependencies are: (i) technical dependency between K-IV Phase 1 (including its transmission line) and K-IV Augmentation; (ii) schedule dependency between KBFU and all K-IV projects, as the former affects the full operation of the latter; (iii) resource dependency between all K-IV projects, given limited public finance; and (iv) information dependency between all K-IV projects. To mitigate these risks, the GoS will notify a K-IV Coordination and E&S Implementation Committee and the KWSC will engage a PMC firm experienced in managing complex capital programs to support program coordination.



VII. RESULTS FRAMEWORK AND MONITORING

PDO Indicators by PDO Outcomes

Baseline	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6	Closing Period
MPA PrDO: Provide safely managed WASH services in urban areas in Pakistan							
MPA PrDO: People provided with safely managed WASH services. (Number of people)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	500,000	4,800,000	8,700,000	14,700,000	17,500,000	17,500,000
Provide safely managed water and sanitation services in Karachi							
People provided with safely managed water (Number of people) ^{CRI}							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	0	7,700,000	7,700,000	16,000,000	16,000,000	16,000,000
➤ People provided with safely managed water - Female (Number of people) ^{CRI}							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	0	3,850,000	3,850,000	8,000,000	8,000,000	8,000,000
➤ People provided with safely managed water - Youth (Number of people) ^{CRI}							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	0	1,386,000	1,386,000	2,880,000	2,880,000	2,880,000
People provided with safely managed sanitation (Number of people) ^{CRI}							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	1,000,000	2,000,000	3,000,000	7,500,000	7,500,000	7,500,000
➤ People provided with safely managed sanitation - Female (Number of people) ^{CRI}							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	500,000	1,000,000	1,500,000	3,750,000	3,750,000	3,750,000
➤ People provided with safely managed sanitation - Youth (Number of people) ^{CRI}							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	180,000	360,000	540,000	1,350,000	1,350,000	1,350,000
Increase KWSC's financial and operational performance							
KWSC Operating Cost Coverage Ratio improved (Percentage)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030



76	80	100	110	120	125	130	130
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Intermediate Indicators by Components

Baseline	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6	Closing Period
Component 1: Capacity Building and Reform of KWSC							
New water and wastewater tariff review framework adopted by KWSC (Yes/No)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	Yes	Yes	Yes	Yes	Yes	Yes
KWSC customer satisfaction improved (Percentage)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	0	60	70	70	80	80
Geographic Information System (GIS) including digital mapping of water supply and sewerage system functional (Yes/No)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
No	No	No	Yes	Yes	Yes	Yes	Yes
KWSC Collection efficiency Improved (Percentage)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
61	63	67	70	73	75	77	77
Water Safety Plan Implemented (Yes/No)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
No	No	Yes	Yes	Yes	Yes	Yes	Yes
Annual appointment of female staff in technical and managerial positions increased (Percentage)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	5	10	15	20	30	30
Gender-inclusive recruitment strategy approved by KWSC board (Yes/No)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
No	No	Yes	Yes	Yes	Yes	Yes	Yes
KWSC staff productivity improved (Number)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
9.50	9.50	8	7	6	5	5	5
Performance agreement signed between KWSC and GoS (Yes/No)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Component 2: Safely Managed Water Supply and Sanitation Services							



Unplanned settlements provided with improved water and sanitation services and hygiene promotion (Number)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	1	2	4	6	7	7
Sewer network rehabilitated or extended (Kilometers)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	10	30	60	80	100	117	117
Water supply distribution network rehabilitated or extended (Kilometers)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	0	50	60	80	110	110
NRW Reduced (Percentage)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
64	64	55	44	30	30	30	30
Specific energy consumption for water supply reduced (kWh per cubic meter) (Number)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0.43	0.43	0.38	0.34	0.32	0.32	0.32	0.32
Bulk water supply capacity increased (Cubic meters/year)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	0	431,400,000	431,400,000	431,400,000	431,400,000	431,400,000
Wastewater Recycled for Industrial Use (Cubic meters/year)							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	0	0	66,000,000	66,000,000	66,000,000	66,000,000
Private capital mobilized through PPPs (Amount(USD))							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	151,000,000	151,000,000	151,000,000	151,000,000	151,000,000	151,000,000
Private capital enabled through PPPs (Amount(USD))							
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Sep/2029	Sep/2030	Dec/2030
0	0	0	0	0	0	0	113,000,000
Component 3: Project Management and Studies							
Component 4: Contingent Emergency Response Component (CERC)							



Monitoring & Evaluation Plan: PrDO and PDO Indicators by PrDO and PDO Outcomes

PrDO: To provide safely managed WASH services in urban areas in Pakistan	
People provided with safely managed WASH services (Number of people)	
Description	Number of people provided with safely managed WASH services under all phases of the program as defined in the World Bank Corporate Scorecard. People must receive all three of the below: Safely managed water means drinking water from an improved water source that is accessible on premises, available when needed and free from fecal and priority chemical contamination. Safely managed sanitation means use of improved facilities that are not shared with other households and where excreta are safely disposed of in situ or removed and treated offsite. Basic hygiene services. Households with a handwashing facility with soap and water available on-premises. Hygiene for reducing transmission and recommend enforcing regular handwashing, ensuring daily disinfection and cleaning of surfaces, providing basic water, sanitation and waste management facilities, and following appropriate environmental cleaning and decontamination procedures. The descriptions of PDO indicators 1 and 2 provide more details.
Frequency	Semi-annual
Data source	PIUs reports, PIUs MISs, Hygiene surveys
Methodology for Data Collection	Details will be in the project operations manual.
Responsibility for Data Collection	PIUs, KWSC zonal teams
PDO: To provide safely managed water and sanitation services in Karachi	
People provided with safely managed water, CRI (Number of people)	
Description	Cumulative number of people provided with safely managed water supply services under the project meeting the following criteria: Improved drinking water sources means piped water or delivered water (tankers). Accessible on premises means house connections or water delivered to household by tankers. Available when needed. For a minimum of 12 hours per day. For piped water, availability data will be collected through pressure loggers in the distribution network. For supply through water tankers, availability data will be collected from KWSC's recently installed digital tanker monitoring system that tracks volume and number of trips. Free from fecal and priority contamination means water supplied is free from E-coli contamination and meets national or WHO standards for arsenic and fluoride. KWSC will institutionalize a system of water quality surveillance at predetermined points in the water distribution network.
Frequency	Semi-annual
Data source	PIU reports, KWSC MIS
Methodology for Data Collection	Details will be in the project operations manual.
Responsibility for Data Collection	PIU, KWSC zonal teams
People provided with safely managed water – Female, CRI (Number of people)	
Description	Cumulative number of females provided with safely managed water supply services under the project
Frequency	Semi-annual
Data source	PIU reports, KWSC MIS
Methodology for Data Collection	As above, beneficiary disaggregated by gender
Responsibility for Data Collection	PIU, KWSC zonal teams
People provided with safely managed water – Youth, CRI (Number of people)	
Description	Cumulative number of youths provided with safely managed water supply services under the project
Frequency	Semi-annual



Data source	PIU reports, KWSC MIS
Methodology for Data Collection	As above, disaggregated by number youth. Youth defined as per UN definition (between 15 to 24 years old).
Responsibility for Data Collection	PIU, KWSC zonal teams
People provided with safely managed sanitation, CRI (Number of people)	
Description	Cumulative number of people provided with safely managed sanitation services under the project, where safely managed sanitation service means: Improved sanitation facility means toilets connected to sewer system or septic tanks. Excreta is removed and treated offsite means connection to a functional sewer network and treatment plant
Frequency	Semi-annual
Data source	PIU reports, KWSC MIS
Methodology for Data Collection	Details will be in project operations manual.
Responsibility for Data Collection	PIU, KWSC zonal teams
People provided with safely managed sanitation – Female, CRI (Number of people)	
Description	Cumulative number of females provided with safely managed sanitation services under the project
Frequency	Semi-annual
Data source	PIU reports, KWSC MIS
Methodology for Data Collection	As above, beneficiary disaggregated by gender
Responsibility for Data Collection	PIU, KWSC zonal teams
People with safely managed sanitation – Youth, CRI (Number of people)	
Description	Cumulative number of youths provided with safely managed sanitation services under the project
Frequency	Semi-annual
Data source	PIU reports, KWSC MIS
Methodology for Data Collection	As above, disaggregated by number youth. Youth defined as per UN definition for age between 15 to 24 years old.
Responsibility for Data Collection	PIU, KWSC Zonal teams
To increase KWSC's financial and operational performance	
KWSC Operating Cost Coverage Ratio improved (Percentage)	
Description	Ratio expressed in percentage of collected revenues for water supply to the total operating cost for water supply but not including debt service or depreciation cost.
Frequency	Annual, within one month of the end of the FY
Data source	PIU and KWSC financial audit reports
Methodology for Data Collection	PIU reports and KWSC entity audits.
Responsibility for Data Collection	PIU

Monitoring & Evaluation Plan: Intermediate Results Indicators by Components

Component 1: Capacity Building and Reform of KWSC
New water and wastewater tariff review framework adopted by KWSC (Yes/No)



Description	Tariff review framework that enables recovery of the full cost of service provision considering the affordability limits of users approved by the responsible authorities and adopted by KWSC board. Full cost of service provision includes production, distribution, and management costs as well as the E&S monitoring and mitigation costs.
Frequency	Once and revised in accordance with the regulatory requirements.
Data source	PIU and KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
KWSC customer satisfaction improved (Percentage)	
Description	Customer Satisfaction Index (CSI) score measuring the level of satisfaction customers have with KWSC's services.
Frequency	Three surveys, first at the end of the first year, second at midterm review and the third - at the project closing.
Data source	Independent customer survey commissioned by KWSC
Methodology for Data Collection	Customer survey questionnaire (based on a Likert-scale of 1 to 7) to assess (i) the importance customers attach to selected service attributes; and (ii) their level of satisfaction on these attributes. The survey will focus on a few service attributes determined through focus groups. Based on the survey, CSI shall be calculated as follows: • average of importance scores for each service attribute (I) • average of satisfaction scores for each service attribute (S) • average importance scores for all service attributes (I_{all}) • calculate weights (W) for each attribute by dividing the average of importance score for each attribute by the average for all attributes, i.e., $W = I / I_{all}$ • calculated weighted satisfaction scores (i.e., satisfaction scores that consider the importance ratings) = $S * W$ • $CSI = \text{average of } S * W \text{ for all service attributes expressed as a percentage.}$
Responsibility for Data Collection	Customer survey firm engaged by KWSC
Geographic Information System (GIS) including digital mapping of water supply and sewerage system functional (Yes/No)	
Description	GIS system is installed, water supply distribution system is mapped, sewerage system is mapped. The system is completed, functional and kept updated.
Frequency	One plan to be monitored Annually
Data source	PIU
Methodology for Data Collection	PIU reports
Responsibility for Data Collection	PIU
KWSC Collection efficiency improved (Percentage)	
Description	Ratio expressed in percentage of collected amounts to the amount billed for the same referenced period in referenced currency. The collected amount may also include accounts receivables and connection/subscription fees.
Frequency	Semi-annual
Data source	PIU and KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
Water Safety Plan Implemented (Yes/No)	
Description	A Water Safety Plan (WSP) is a risk assessment and management tool that aims to ensure the safety of a drinking water supply. An implemented plan will be annually updated by KWSC. An implemented plan is adopted and testing and monitoring of water quality is carried out as per the plan.
Frequency	Once and update as necessary
Data source	KWSC



Methodology for Data Collection	PIU and KWSC Reports
Responsibility for Data Collection	KWSC
Annual appointment of female staff in technical and managerial positions increased (Percentage)	
Description	Percentage of women hired out of the total newly appointments in technical and managerial position in that year (i.e. the total number of women hired / total number of people hired in technical and managerial positions).
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
Gender-inclusive recruitment strategy approved by KWSC board (Yes/No)	
Description	Gender inclusive strategy developed and adopted by KWSC with guidance to promote female staff.
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC Reports
Responsibility for Data Collection	PIU
KWSC staff productivity improved (Number)	
Description	Total number of KWSC staff divided by number of thousand registered customers.
Frequency	Annual
Data source	KWSC
Methodology for Data Collection	KWSC Customer Digitalization Reports
Responsibility for Data Collection	KWSC
Performance agreement signed between KWSC and GoS (Yes/No)	
Description	A utility performance-based agreement that sets out role, scope, and KWSC performance targets and review mechanism, satisfactory to the Bank is prepared and signed between the KWSC and the GoS
Frequency	3-years contract to be monitored annually
Data source	PIU and KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
Component 2: Safely Managed Water Supply and Sanitation Services	
Unplanned settlements provided with improved water and sanitation services and hygiene promotion (Number)	
Description	Katchi Abadis provided with improved water and sanitation services and hygiene promotion through behavior change campaigns
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU



Sewer network rehabilitated or extended (Kilometers)	
Description	Length of sewerage network rehabilitated and extended under the project.
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
Water supply distribution network rehabilitated or extended (Kilometers)	
Description	Length of water distribution network rehabilitated in Katchi Abadis under the project
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
NRW Reduced (Percentage)	
Description	The percentage of NRW measured as billed water quantity to supplied water quantity within the stipulated period. The NRW will account for both commercial and leakage losses.
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
Specific energy consumption for water supply reduced (kWh per cubic meter) (Number)	
Description	Electricity consumption at the pump stations divided by the total quantity of water entering the supply system
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
Bulk water supply capacity increased (Cubic meters/year)	
Description	Bulk water supply capacity developed through the K-IV Augmentation scheme
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
Wastewater Recycled for Industrial Use (Cubic meters/year)	
Description	Recycled wastewater meeting quality standards for industrial use delivered and paid for by industrial users.
Frequency	Semi-annual
Data source	KWSC
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU



Private capital mobilized through PPPs (US \$)	
Description	Private capital invested in wastewater and water treatment attributed to World Bank financing during project duration.
Frequency	Upon PPP financial close
Data source	KWSC PPP contracts
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU
Private capital enabled through PPPs (US \$)	
Description	Amount of private capital investments attributed to World Bank financing made in KWSC projects within three years of project closure through PPPs in bulk water and water treatment plants
Frequency	Upon PPP financial close and if applicable, at each Implementation Status and Results Report (ISR), Implementation Completion Report (ICR), and 3 years post project closure
Data source	KWSC PPP contracts
Methodology for Data Collection	PIU and KWSC reports
Responsibility for Data Collection	PIU



ANNEX 1: Implementation Arrangements and Support Plan

COUNTRY: Islamic Republic of Pakistan

Second Karachi Water and Sewerage Services Improvement Project
Pakistan Urban Water, Sanitation, and Hygiene Services (WASH) Services Program

A. Project Institutional and Implementation Arrangements

1. **The Implementing Agency of this project will be the GoS through the KWSC and the PIU established within the KWSC to manage KWSSIP-1 and subsequent KWSSIP projects.** The PIU will report to KWSC management represented by the Chief Executive Officer/Managing Director and maintain liaison with the Board of Directors. The PIU and associated consultants are expected to reach 50 percent women's participation over the project life.
2. **Performance Agreement Assessment.** The KWSC will select and manage a firm tasked with independent verification of the KWSC's Performance Agreement, following terms of reference satisfactory to the Bank. The firm will verify achievement of targets, and in doing so shall examine secondary data, collect primary data, and conduct other measurements annually or periodically as needed. The firm will share the findings with the KWSC, who will, in turn, share them with the GoP and the Bank.
3. **A PSC has been established to provide oversight and high-level coordination.** The PSC includes high-level representatives of all agencies involved in the PSC. It provides planning and strategic guidance and facilitates multi-stakeholder cooperation for the development and implementation of a vision, policy reforms, and investment program towards the improvement of Karachi's WASH services. The PSC has defined responsibilities to minimize overlaps, mitigate potential disagreements with the KWSC, and will be empowered to evolve its membership in case of changing circumstances.
4. **Pursuant to a Project Co-Lenders' Agreement with the AIIB, (which will be very similar to the one for KWSSIP-1), the World Bank will supervise the project and administer both the IBRD and the AIIB loans in accordance with the World Bank's policies and procedures.** The co-lender agreement will be signed by the World Bank and the AIIB in accordance with the 2016 Co-Financing Framework Agreement between World Bank and the AIIB (as amended in 2018). The AIIB and the World Bank will jointly finance the project in equal shares, with all components financed jointly by the partners.

B. Financial Management and Disbursements

5. **The project has a residual FM risk rating of Moderate (subject to timely implementation of the recommended mitigation measures and Action Plan) and has met the minimum FM requirement in accordance with the requirements of Bank Policy for Investment Project Financing effective October 1, 2018.** The FM arrangements provide reasonable assurance that the financing proceeds will be used for the intended purposes, with due attention to the principles of economy, efficiency, effectiveness, transparency, and accountability to support implementation and achieve the desired results. This objective will be sustained by ensuring that strong FM systems are maintained for the project throughout implementation. Detailed FM reviews will be conducted regularly, either within the proposed supervision plan or on a more frequent schedule if needed, to ensure that expenditures incurred by the project remain eligible.
6. **Disbursements will be report-based with advance equivalent to six months forecast of expected**



payments. A segregated DA will be established following the Revolving Fund Assignment Account procedures by the Finance Division, GoP. Eligible expenditures will be comprised of works, consulting and non-consulting services, goods, workshops and training, and incremental operating costs incurred for implementing the project.

7. **Risk Assessment and Mitigation.** Table A1.1 identifies the key risks and required mitigation measures. The following are necessary features of a strong FM system:

- An adequate number and mix of skilled and experienced staff in the accounting unit.
- The internal control system should ensure the conduct of an orderly and efficient payment and procurement process, and proper recording and safeguarding of assets and resources.
- The accounting system should support the project's requests for funding and meet its reporting obligations.
- An internal audit firm will review compliance with the rules and the project's financial statements and internal controls. The firm already hired for the KWSSIP-1 will perform this function for the KWSSIP-2 as well.

Table A1.1: Risk Assessment and Mitigation

Risk	Risk Rating	Risk Mitigating Measures Incorporated into Project Design	Residual Risk Rating
Country Level	Substantial		Substantial
Entity Level	Substantial	Capacity of FM unit needs enhancement and the requisite staff should remain available.	Moderate
Project Level	Substantial	As above.	Moderate
Control Risk			
Budgeting	Moderate	Detailed budget and a Project Plan with quarterly breakup of activities along with estimated funds utilization report will be prepared no later than one (1) month after the Effective Date.	Low
Accounting	Moderate	Activity-wise expenditure recorded using the existing chart of accounts.	Moderate
Internal Control	Substantial	Project team will follow the GoS rules and regulations along with the project operations manual for conduct of activities. Internal Audit function will be strengthened by employing the services of an Internal Audit Firm.	Moderate
Fund Flow	High	FM standard operating procedures to clearly define the funds flow arrangements and service standards for release of funds. GoS to ensure timely provision of counterpart funds to avoid delays in project operations.	Substantial
Financial Reporting	Moderate	FM trainings to be imparted to staff to provide handholding and on-field technical support.	Moderate
Auditing	Low	Auditor General of Pakistan conducts audit of the project as per International Organization of Supreme Audit Institutions Standards which are acceptable.	Low

8. **FM Staffing.** The project shall retain the FM Specialist position from KWSSIP-1 and also hire a junior FM Specialist to support the increased workload under KWSSIP-2. This unit will also perform the FM function of KWSSIP-2.

9. **Budgeting and Planning.** The project is part of the Annual Development Program and is reflected



in the GoS budget. The annual budget for the project will be prepared by the PIU, which will be responsible for the preparation of budgets for all subprojects based on planned activities identified in the procurement plan and will be timely communicated to the Finance Manager for the allocation of resources for counterpart funding and the annual budget of the project. The Budget Section will seek input from all relevant departments and the Project Director for the preparation of the budget.

10. **Accounting and Financial Reporting.** The project accounts will be maintained on a cash basis as per GoS accounting policies and procedures and using the existing Chart of Accounts, which is comprehensive. Payment vouchers will be prepared for each transaction, and the relevant accounting codes, disbursement category, and project component will be mentioned on the payment vouchers.

11. For each semester, KWSSIP-2 will prepare a half-yearly IUF that will be furnished to the Bank within 45 days of the close of the semester. IUFRs will report funds receipt and utilization during the quarter according to project components and disbursement categories. In addition, FM staff will prepare annual project financial statements as per the requirements of International Public Sector Accounting Standards Cash Basis and submit audited project financial statements within six months of close of financial year.

12. **Internal Controls.** Project documents describe standard processes, minimum documentation, and approval limits for processing different types of payments. Internal control activities at the project will, at a minimum, include the authorization and approval activity from the relevant authorities; segregation of duties; regular process verification of FM processes; ensuring physical controls for the assets; and effective internal audit to be performed twice a year by a firm of chartered accountants.

13. **Fund Flow Arrangements.** The project will mainly use the Advance and Direct Payment method of disbursements in line with the Disbursement and Financial Information Letter. For receipt of funds, the Implementing Agency will open a segregated DA in US Dollars at the National Bank of Pakistan. The DA will be operated in accordance with the provisions of "Revised Accounting Procedure for Revolving Fund Account (Foreign Aid Assignment Account)" issued by the Finance Division.

14. **Disbursements will be report-based, and the project will mainly use the advance method of disbursement where the funds will be front-loaded into the DA based on six months cash forecast.** Initial advance into DA will be provided by the Bank based on projections for the first six months. Subsequent advances will be based on forecasts for the following six months. The expenditure incurred during a semester against advance will also be documented in the Bank's system, called *Client Connection*, based on the IUFRs. Further details regarding disbursements are provided in the Disbursement and Financial Information Letter. At the end of the project, if any unspent funds are lying with the Implementing Agency, the same will be refunded to the World Bank. In addition to the advance method, other methods of disbursement, which include direct payment and reimbursement, will be used as and when needed.

15. **Auditing.** Project financial statement will be audited by the Auditor General of Pakistan through the Director General Audit Sindh. Audited financial statements would be submitted to the Bank within six months of the close of each FY.

16. **Implementation Support Plan.** Two annual field supervision missions will review the FM arrangements at the PIU. The internal and external auditors' reports, as well as the project's supervision reports, would be the primary sources of information about the FM arrangements at KWSSIP-2.

C. Procurement

17. Procurement arrangements at the PIU will continue; however, additional staff will be required



for technical support, procurement, and contract management. Evaluation committees will be formed within the PIU, and the Project Director will approve all procurement decisions and sign all contracts. The PPP node is agreed to be developed within the KWSC; however, the decision-making process regarding the PPP node and PPP unit of GoS is to be finalized. The PPSD will be updated on the PPP implementation aspects, including the type of standard procurement documents to be used to develop PPP documents for the planned interventions.

18. The identified risk and mitigation measures are summarized in Table A1.2.

Table A1.2: Risks and Mitigation Measures

<i>Risk Area</i>	<i>Risk Description</i>	<i>Mitigation Measure</i>
Delayed decision-making	There could be delays in decision-making, specifically on large contracts and PPP interventions. Decision-making on PPP interventions after the selection of bidding documents may require discussions within the various stakeholders (PIU, PPP unit, or even the PSC).	Project Director will continue to have delegation of powers in decision-making on evaluation and award of contracts as well as contract management. Clarity has been obtained from Sindh Planning and Development Department that there will be no need of re-approval if cost variation of any individual contract is up to 15 percent and the cumulative cost of all contracts does not exceed the federal government's appraisal document ceiling amount. Clarity will be obtained on PPP decisions by negotiations.
Inadequate market capacity	Market participation could be a challenge for the largest contracts of K-IV Augmentation works from international firms; other works contracts on National Competitive Bidding and smaller International Competitive Biddings; Consultancy contracts of design and supervision and project management support. PPP interventions are also sensitive to market participation.	K-IV Augmentation will be procured as one contract of three packages after pre-qualification to obtain economies of scale and attract the largest firms active in the region. For the other contracts, packaging will be optimized. For all works contract price adjustment clause will be provided to limit the bidders' risk. Selection process of consultancy of K-IV design review and construction supervision has commenced; however, the Bank's selection procedure is being used to ensure good participation. Selection process for design and supervision contract of works other than K-IV Augmentation (Katchi Abadis, NRW, etc.) and project management firm will commence as early as possible enabling adequate support to the PIU. Feasibility studies of PPP interventions will address the technical and financing models, revenue mechanisms, contracting modalities, and concession terms. Potential market will also be assessed before these interventions are launched.
Delays in evaluation and contract awards	Delays in contract awards are possible due to reasons such as evaluation delays, approval delays, less than optimal specification /terms of reference, revision of	The current PIU competencies on technical and procurement aspects will be enhanced immediately, with additional hirings at market competitive scales. Evaluation committee members will be adequately qualified, and additional procurement training will be provided by the Bank. Subject experts will be coopted in committees as/if



	bidding documents and complaints.	needed. Procurement timeline will be agreed for the evaluation and approval process within the PIU. The committees will be involved consistently in the preparation of procurement documents as well as evaluation and recommendation for award. Complaints will be handled promptly and responded to within the stipulated time. The design and supervision firms will be hired on priority so that adequate support is provided in evaluations.
Cost and time overruns and contract administration	Cost and time overruns are possible, which will impact project success.	Cost estimates will be prepared based on market rates. PIU contract manager(s) will be appointed for each contract and procedures will be agreed and documented to allocate responsibilities for timely identification of issues. Contract management, Key Performance Indicators, and monitoring plans will be developed for large value contracts.
Import delays and restrictions	Import of material and foreign currency limitations in the country may create supply delays.	Procurement plans (run up time and approach) will be prepared considering possible constraints in imports due to foreign currency payment restrictions to local firms. Relevant Incoterms will be used in bidding documents to facilitate bidding and payments.
Inadequate design and implementation of PPPs	There could be a tendency to commence PPP activities without comprehensive technical and financial feasibility, financing /revenue model and market participation.	Adequate PPP implementation arrangements will be agreed within the PIU. Robust project assessment will be ensured finalizing the available studies and filling in the gap if any. Resource allocation will be ensured for structuring the interventions, developing contracting modalities, facilitating healthy participation, and contract management. PPP fiscal risk assessment (PFRAM) to be undertaken and consideration of measures to address arising fiscal contingency risks, for example partial risk guarantees.